

Recent advances and future perspectives on key components for sodium-ion batteries

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ABSTRACT

Sodium-ion batteries (SIBs) have emerged as a promising alternative to lithium-based technologies, especially for large-scale energy storage, due to high safety, environmental sustainability, abundant resource and lower cost. However, SIBs performance still needs to be improved in the areas of cyclic stability, energy/power densities and fast-charging capability. This work provides a comprehensive analysis of SIBs, covering their basic function, recent advancements in materials, and crucial components such as anodes, cathodes, electrolytes, and separators. First, the historical development, working principle and electrode materials of SIBs are symmetrically summarized. Second, the recent progress of cathode materials such as polyanionic compounds, Prussian blue, layer transition metal oxides, and organic compounds are explained. Then, the five primary categories of anode materials for SIBs such as alloying, intercalation, organic, conversion and MXene materials are critically reviewed. In order to develop high performance SIBs for real-world applications, the technical difficulties are finally outlined, and potential future research directions for resolving these issues are suggested. This comprehensive review of the key components of SIB technology contributes important insights given the need of shifting to more sustainable energy storage technologies to address climate change and accomplish the Sustainable Development Goals. It emphasizes how important the key components are to the development of SIBs as high energy/power densities and environmentally friendly energy storage options.

1. Introduction

The most popular energy source in the world is fossil fuels [1]. Numerous intermittent renewable and cleaner energy sources, including wind, solar, and wave power, have rapidly grown because of the risks associated with resource depletion, environmental pollution, and political instability surrounding the production of fossil fuels [2,3]. A large-scale energy storage system (ESS) is essential to peak shift operation in order to integrate these renewable energies into the electrical grid [4]. Because of its adaptability, high energy conversion efficiency, and ease of maintenance, electrochemical secondary batteries are one of the most promising energy storage technologies for large-scale

electricity storage [5]. Therefore, it is essential to build battery systems that are robust, economical, high-performing, ease of maintenance, high energy conversion efficiency, flexibility and environmentally friendly. Hence, rechargeable batteries are the most promising energy storage technologies for large-scale electricity storage [6]. Because of their high energy density and extended cyclic life, rechargeable lithium-ion batteries (LIBs) have been commercialized and have taken on a significant role as secondary batteries [7]. However, the price of lithium is rising due to the unequal distribution of lithium resources and numerous ongoing consumptions [8]. Therefore, it is important to develop low-cost alternative batteries for large scale application. Sodium-ion batteries (SIBs) are evolving as a potential alternative to lithium-ion batteries

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(LIBs) in the quest for affordable and environmentally friendly energy storage options [9]. The most promising alternative ion battery technology is sodium-ion batteries (SIBs), which have comparable cell designs and materials and operate on the same principles as lithium-ion batteries (LIBs) [10]. SIBs are attractive complementary power sources for electric vehicles and energy storage systems since they can be mass-produced using the same manufacturing infrastructure used for LIBs [11]. The most significant benefit of SIBs is that sodium is inexpensive because of its high abundance in the Earth's crust (23,000 ppm, or 2.3%), which is more than a thousand times more prevalent than lithium (20 ppm) [12]. With the exception of their ion carriers, SIBs and LIBs have nearly identical battery components and electrical storage mechanisms. Since sodium and lithium have relatively similar intercalation chemistry, identical materials can be used for both systems when it relates to cathode materials [13]. Nonetheless, there are a few clear distinctions between these systems. Phase stability, transport characteristics, and interphase formation are influenced by the bigger Na^+ ions (1.02 Å) compared to Li^+ ions (0.76 Å). SIBs will always fall short in terms of energy density since sodium is heavier than lithium (23 g mol⁻¹ against 6.9 g mol⁻¹) and has a larger standard electrode potential (−2.71 V vs. SHE versus −3.02 V vs. SHE for lithium) [14]. However, the capacity is mostly governed by the properties of the host structures that operate as electrodes, and the weight of cyclable Li or Na is a tiny portion of the mass of the components. Therefore, the shift from LIBs to SIBs should, in theory, have no effects on energy density. Additionally, aluminum and lithium undergo an alloy reaction below 0.1 V vs. Li/Li^+ , indicating that aluminum can be used as a current collector for sodium cell anodes. As a result, aluminum is a more affordable option for SIB anode current collectors than copper [15]. Both the anode and cathode materials are the most essential components of SIBs to facilitate the electrochemical process. Moreover, it also depends on electrolytes, binders, conductive agents, and separators [16]. Research on cathode materials has advanced rapidly in recent years, leading to the discovery of several novel materials with superior qualities, including high-entropy materials [17]. Cathode materials are typically developed as cobalt-free materials in an effort to reduce environmental pollution [18]. Prussian blue analogues (PBAs), organic compounds, layer transition oxides, and polyanionic compounds such as sulphates, pyrophosphates, fluorosulfates, and phosphates are the main categories of these cathode materials [19].

The metal type of sodium anodes, the conversion reaction type (metal sulfides, phosphides, metal oxides, metal etc.), the alloying reaction type (silicon, tin, phosphorus, antimony, germanium, etc.), and the insertion reaction type (carbon-based materials and titanium oxides) are the four main categories of SIB anode materials based on the charge/discharge reaction mechanisms [20]. In recent years, a few novel kinds of anodes have been created. These novel anode materials have numerous active sites and multiple reaction routes. It is important to note that metal–organic frameworks (MOFs) have received increasing attention in recent years as SIB anodes [21]. MOFs can offer several active sites because they are inorganic organic hybrids. Metal nodes and organic ligands can both operate as active sites to offer particular capacities [22,23]. Nevertheless, there is currently a lack of thorough research on the reaction mechanism of MOFs as anode materials [24,25]. Hence, the development of SIBs is still hampered by the demand to find an anode with appropriate Na voltage storage, a large reversible capacity, and good structural stability. A common anode material in LIBs, graphite has a moderate Li storage capacity of about 350 mA h g⁻¹ at about 0.1 V vs. Li/Li^+ . Graphite does not appropriately intercalate sodium ions, as recent research has shown [8]. Sodium ions can be inserted into non-graphitic anodes, which are mostly made of other carbonaceous materials like carbon black and pitch-based carbon fibers. Hard carbons have been widely and thermally tested characterized and modeled in Na cells, which are produced at high temperatures from carbon-based precursors. For SIB systems, these non-graphitic carbonaceous materials are regarded as the preferred “first-

generation” anodes. Due to dendritic development, high reactivity, and an unstable passivation layer in the majority of organic electrolytes at normal temperature, sodium metal is not used to fabricate SIBs [26]. Even more challenging than Li metal anodes are the strong reactivity of metallic sodium with organic electrolyte solvents and dendritic development during Na metal deposition [27]. Devices using Na metal electrodes at room temperature are also at risk due to sodium's low melting point of 97.7 °C. Therefore, it is crucial to choose a real Na-ion system, in which Na ions are transported in a “rocking-chair” arrangement between cathodes and anodes [28].

Moreover, electrolytes, conductive agents, binders, and electrode materials are all essential for improving SIBs [29]. Additionally, new kinds of adding agents are investigated to enhance electrochemical performance [30]. Acetylene black (AB), ketjen black (KB) and Super-P are common conductive conductors. Polyvinyl alcohol (PVA), carboxymethyl cellulose (CMC), polyacrylic acid (PAA), and Polyvinylidene fluoride (PVDF) are the primary SIB binders [31–33].

The number of publications per year about sodium batteries has significantly increased over the last ten years, as seen in Fig. 1. However, SIB performance still needs to be improved in the areas of cycle stability, energy/power densities, and rapid charging capability [34]. Compared to LIBs, SIBs have a lower energy density, which is their main drawback in energy storage. [35]. This work provides a balanced strategy by examining both limitations and the innovative approaches emerging for addressing them, in contrast to previous reviews that frequently highlight the challenges associated with SIBs. Additionally, this review aims to draw attention to understudied areas of SIB research such as the potential of novel electrolyte systems to improve ionic conductivity and stability, and the role of novel cathode and anode materials in improving battery performance. This article aims to provide a comprehensive resource for scholars, policymakers, and industry stakeholders by summarizing the most recent research findings and identifying emerging trends.

2. Historical development and working principle of SIBs

The historical development of SIBs and LIBs is comparable. Since 1991, Sony has started marketing LIBs. Driven by the desire for rechargeable energy storage technologies, research on SIBs and early LIB development started in the 1970 s [36]. The discovery of stable sodium transition metal oxides in the 1980 s renewed interest in sodium-based systems and broadened the field of positive electrode (cathode) chemistries, even though LIBs swiftly surpassed SIBs because of their higher energy density and electrochemical stability [37]. After the mid-2000 s,

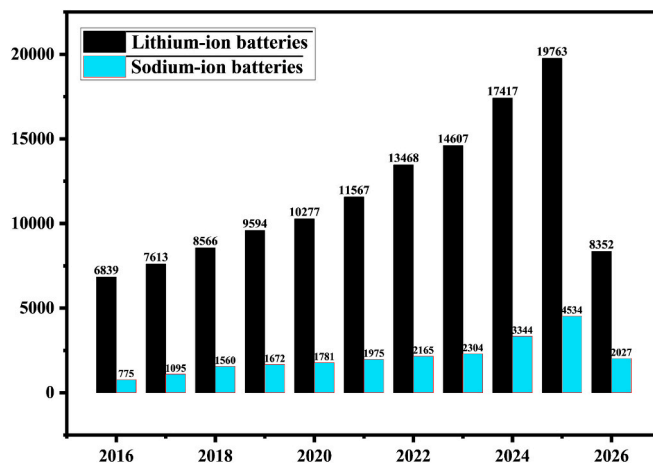


Fig. 1. Comparison of number of publications of Na-ion battery and Li-ion battery over the last ten years using Scopus database (last updated on the 4th May 2026).

SIBs gained momentum due to the identification of hard carbon (HC) as a feasible negative electrode (anode) material, which laid the groundwork for the creation of workable complete cell designs [38]. LIBs are now the primary energy storage technologies used in consumer electronics, the transportation industry (such as electric cars), the stationary energy storage industry (such as datacenter backup power), and other uses [39]. Because sodium-ion batteries have a lower energy density than LIBs, there was a decrease in research between 1990 and 2000. Industry interest in SIBs grew by 2010, resulting in pilot-scale projects and commercial development (Fig. 2(a)) [40]. However, the development of sodium-ion technology was accelerated between 2010 and 2022 since sodium-ion batteries are seen as the next generation of environmentally friendly and inexpensive energy storage technologies [41]. The shortages of cobalt and lithium, which are essential for battery manufacture, may worsen in the upcoming years due to the growing number of proposed gigafactories and production capacity [42]. This makes SIBs an attractive substitute for applications that prioritize resource security, scalability, and cost over optimizing gravimetric energy density. Different kinds of cathode materials such as prussian blue analogues

(PBAs), polyanionic compounds, and layered oxides have emerged from early materials research [43]. The trade-offs between capacity, operating voltage, and structural stability are different for each class. For example, PBAs show good rate capability and rapid sodium-ion diffusion, whereas polyanionic compounds provide improved operating voltages and thermal stability (Fig. 2(b)). HC continues to be the most economically feasible and electrochemically stable material on the anode side [44]. Lead, antimony, and tin are examples of alloying- and conversion-type anodes that have higher theoretical capacities, but their practical use is hampered by significant volumetric changes during cycling, which result in quick capacity fade and short cycle life [45]. SIBs currently show lower energy densities than state-of-the-art LIBs due to their lower average cell voltages and the higher atomic mass of sodium-based materials [46]. These variations stem from basic physicochemical differences between lithium ions (Li^+) and sodium ions (Na^+). Na^+ has a lower standard redox potential, an average 28% lower desolvation energy, and an ionic radius that is around 34% greater. In comparison to LIBs, these characteristics collectively result in slower solid-state diffusion kinetics, difficulties generating stable solid-

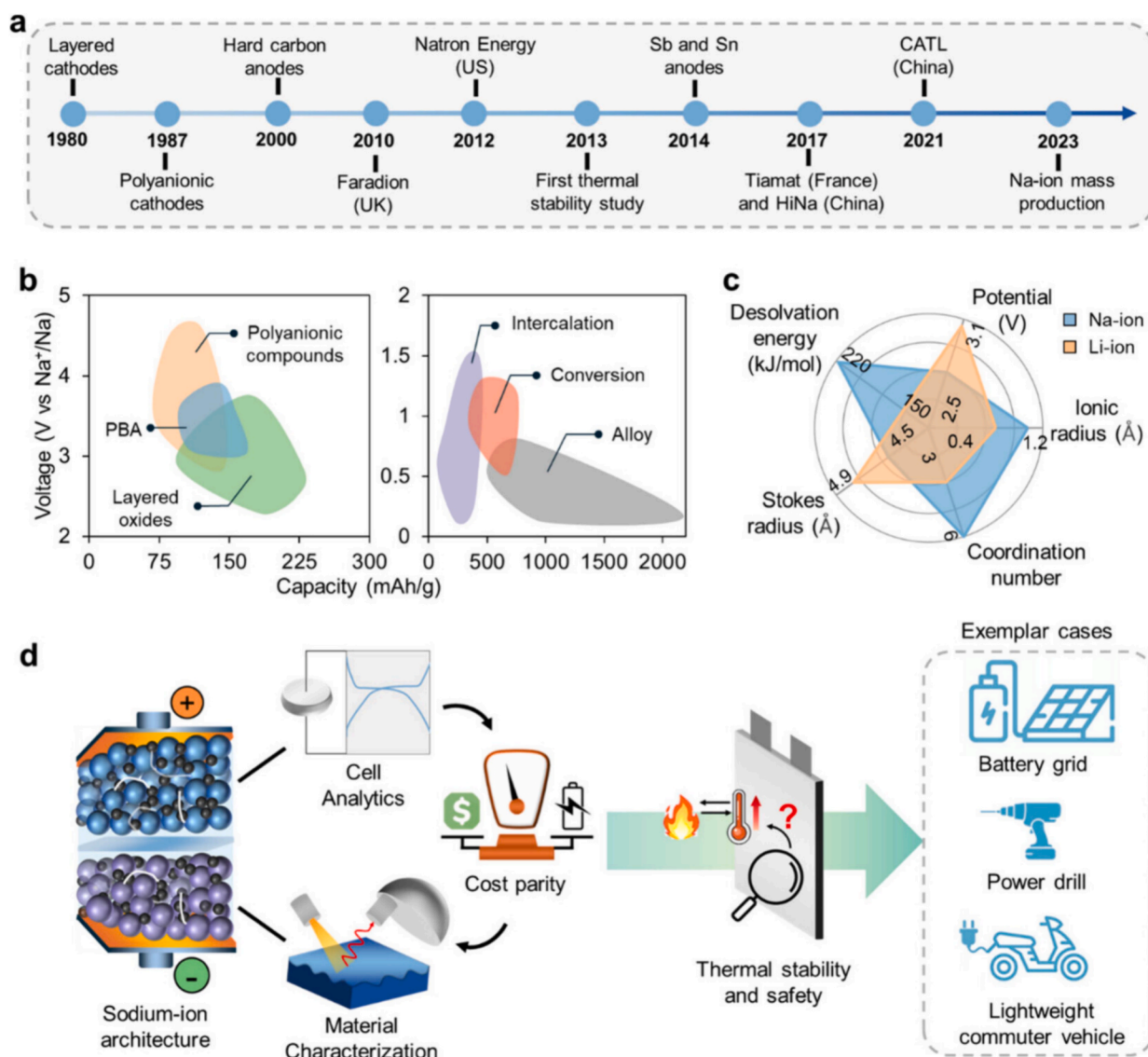


Fig. 2. (a) A chronology of the development of sodium-ion batteries (SIBs) that highlights significant advancements in cathode and anode chemistries as well as the advent of new commercial and mass-manufacturing initiatives. (b) A comparison of different cathode and anode materials' energy storage capabilities. (c) A comparison of the salient features of lithium and sodium ions. (d) A schematic summary of the important factors to take into account when creating useful SIBs. Reproduced under CC BY-NC-ND 4.0 license, with permission from ref [40]. Copyright © 2025 ACS.

electrolyte interphases (SEIs), and a decrease in feasible cell voltage (Fig. 2(c)). Advances in electrode design, electrolyte composition, and cell architecture, along with cost concerns, are crucial for facilitating the wider commercial use of SIBs, as shown in Fig. 2(d). The most common type of electrolyte is still nonaqueous liquid electrolytes that contain solvents (linear and cyclic carbonates, ethers, etc.), salts (NaClO_4 , NaBF_4 , NaPF_6 , etc.), and specific additives (flame retardants, film formers, etc.).

During charging, lithium ions move via the electrolyte from the cathode to the anode, as illustrated in Fig. 3a, and during discharging, they do the opposite. A conventional LIB consists of four basic parts: an electrolyte, an anode, a separator, and a cathode. Li-ions can be transported while direct electrode contact and short circuits are avoided due to the separator, a porous insulating layer. With the exception of their ion carriers, the battery components and electrical storage method of SIBs and LIBs are essentially the same. Since the intercalation chemistry of sodium and lithium is so similar, identical compounds can be used for both systems in terms of cathode materials [47]. However, there are some obvious differences between these systems. The larger Na^+ ions (1.02 Å) in comparison to Li^+ ions (0.76 Å) affect transport properties, interphase formation and phase stability (Fig. 2(c)) [48]. SIBs will always fall short in terms of energy density because sodium has a higher standard electrode potential (−2.71 V vs. SHE compared to −3.02 V vs. SHE for lithium) and is heavier than lithium (23 g mol^{−1} compared to 6.9 g mol^{−1}) [49]. However, the capacity is mostly dictated by the properties of the host structures that operate as electrodes, and the weight of cyclable Li or Na is a tiny portion of the mass of the components [50]. SIBs do not alloy with aluminum at low potentials, in contrast to LIBs. This makes it possible to replace the heavier copper foil needed for LIB anodes with lightweight aluminum current collectors on both the cathode and anode. The energy density drawbacks of the sodium chemistry itself may be partially or completely compensated for by this mass reduction at the device level [51]. Because the materials directly affect the battery's energy density, safety, lifespan, cost, and many other aspects, every component of the SIB cell is essential to the battery's overall performance. As shown in Fig. 3b, the basic cell is made up of an electrolyte, a separator, current collectors, a positive electrode, and a negative electrode [52]. Similar to LIBs, SIBs rely on the reversible transfer of ions between the positive and negative electrodes to store or release energy. The primary distinction is that Na ions are employed as the charge carriers rather than Li ions. Sodium ions at the negative electrode release an electron during the discharge process, giving them a positive charge (Na^+). The electrolyte subsequently carries the Na^+ ions to the positive electrode [53]. Meanwhile, the liberated electrons move through an external circuit and a current collector, producing a current that can power devices. Electrons in the positive electrode are forced to travel in the opposite direction toward the negative electrode through

the external circuit when an external voltage is provided during the charge. Concurrently, the Na^+ ions return to the negative electrode via the electrolyte and recombine with the electrons.

3. Components and electrode materials of SIBs

Recently, various alkali metal-ion batteries, such as those based on zinc (Zn^{2+}), magnesium (Mg^{2+}), potassium (K^+) and sodium (Na^+) have attracted attention in an attempt to partially replace lithium-ion batteries (LIBs) [49]. Because of its advantageous electrochemical performance characteristics, which are similar to those of LIBs, and the fact that sodium (2.83 wt%) is much more abundant on Earth than lithium (0.0065 wt%), SIBs became the leading candidates for energy storage devices [55]. In addition to making SIBs a more sustainable choice, this availability also lessens supply chain issues related to lithium supplies. The energy densities of current SIB prototypes are 90–150 Wh/kg, which is still less than the 130–285 Wh/kg that LIBs normally reach. Though they still fall short of higher-energy chemistries like Nickel Manganese Cobalt (NMC) and Lithium Nickel Cobalt Aluminum Oxide, these values are equivalent to LiFePO_4 -based LIBs, which represent the lower-energy end of lithium systems. While NMC batteries typically provide between 1000 and 2000 cycles, SIBs have demonstrated remarkable cycle life, sometimes approaching 4000 cycles (e.g., $\text{Na}_3\text{V}_2(\text{PO}_4)_2\text{F}_3$ combined with hard carbon in the ALISTORE prototype) [56]. Lithium is the best material for high-performance batteries because it has the highest gravimetric and volumetric capacities (3829 mAh/g and 274 mAh/g), low dissolution energy, and good ionic mobility. However, its cost is expensive. Although their bigger ionic sizes and lower voltages can limit energy density, sodium and potassium offer far smaller capacities but have lower costs, greater availability, and higher recyclability. Compared to lithium, sodium has a more noticeable standard electrode potential (2.71 V versus 3.02 V). As a result, researchers have concentrated on using targeted materials engineering to overcome the energy density constraints of SIBs while preserving or increasing their cycle life (Fig. 4) [57]. Electrolytes, separators, binders, and cathode and anode materials are the main constituents of SIBs [49]. By separating the cathode and anode, the separator keeps potential short circuits from happening. Additionally, they allow sodium ions to migrate freely between the cathode and anode, serving as carriers that facilitate the processes of energy release and storage [58]. Additionally, SIBs use inexpensive aluminum foil as the current collector because aluminum does not react with sodium, making it suitable for both the anode and cathode current collectors [59].

SIBs function via a rocking-chair mechanism, just as lithium-ion batteries (LIBs). As a result, sodium ions move from the anode to the cathode for storage during the discharge phase and back to the anode during the charging phase. As a result, sodium storage reactions must be

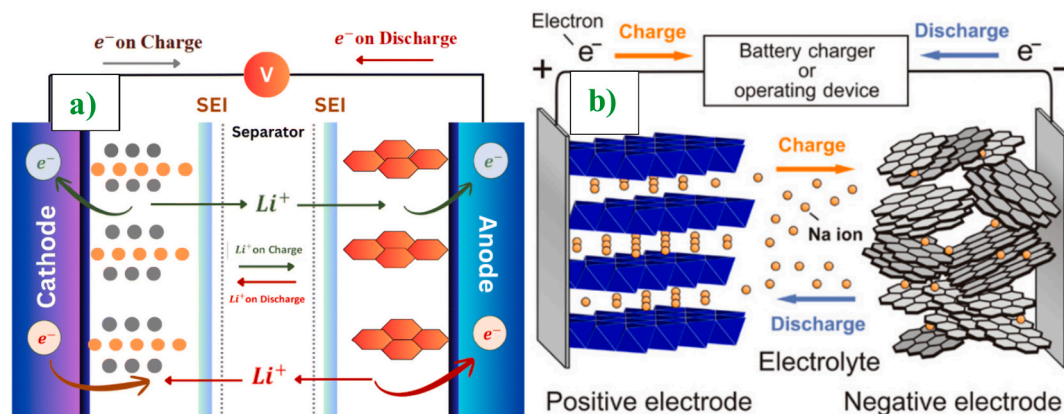


Fig. 3. Working principle of: a) LIBs, reproduced under Creative Commons Attribution-Non-Commercial 3.0 Unported Licence, with permission from ref. [54]. Copyright © 2025 RSC. b) SIBs, reproduced under the CC BY-NC-ND 4.0 license, with permission from ref. [52]. Copyright © 2025 MDPI.

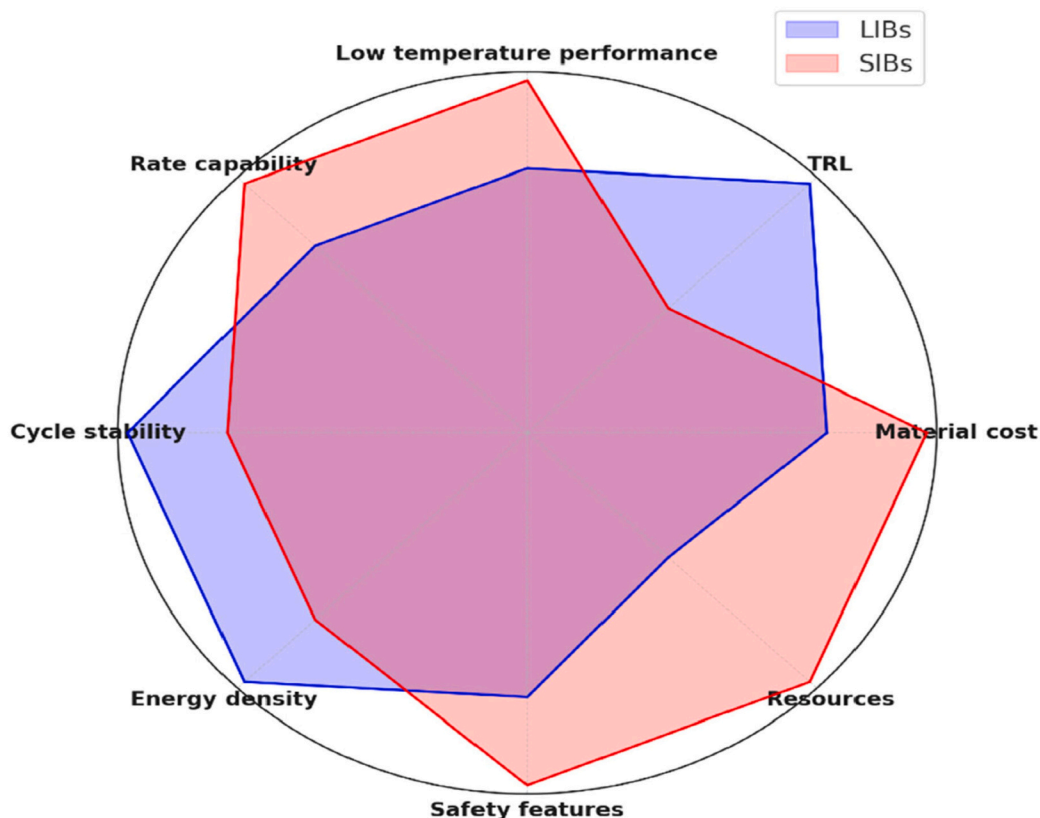


Fig. 4. Comparison of SIBs and LIBs based on different battery features. Reproduced under the CC-BY-4.0 licence, with permission from ref. [60]. Copyright © 2015, Wiley.

possible in both cathode and anode materials, with a possible distinction between them [61]. The battery's capacity to charge and discharge is improved by a greater potential difference. Because of this, LIBs and SIBs are quite similar, but the bigger size and higher redox potential of Na^+ result in inevitable drawbacks. For example, the graphite anode of LIBs,

which is frequently employed, does not significantly intercalate sodium [62].

However, SIBs are being considered as a replacement for LIBs due to the scarcity of lithium resources and the abundance of sodium resources; therefore, finding an appropriate anode material for SIBs is necessary.

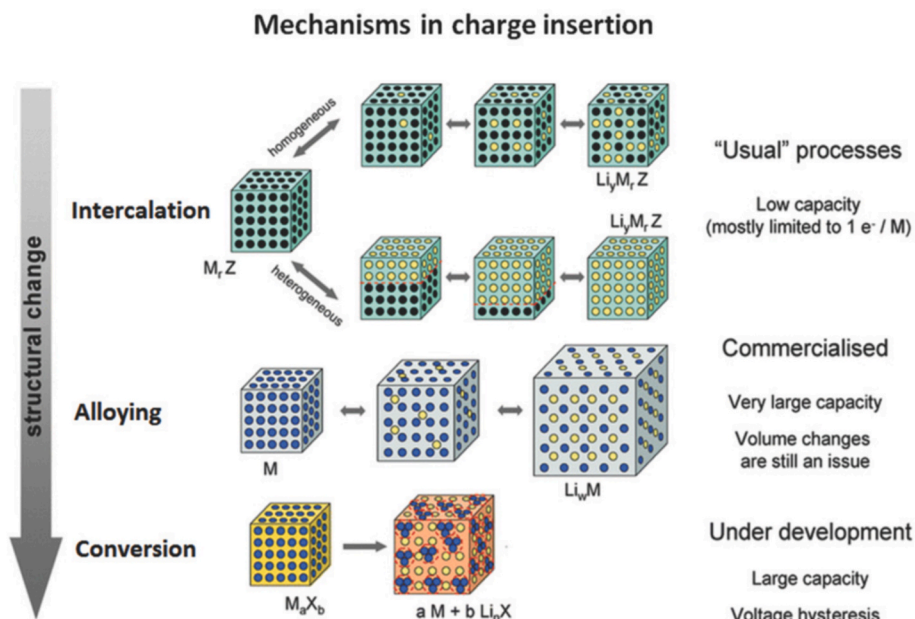


Fig. 5. An illustration of the various charge insertion processes seen in lithium-ion battery electrode materials. Blue circles indicate metal, yellow circles indicate lithium, and black circles indicate gaps in the crystal structure. Sodium-ion batteries are classified similarly. Reproduced with permission [10]. Copyright 2018, Wiley-VCH.

Intercalation, alloying, and conversion are three types of charge insertion methods that allow sodium to be inserted into the matrix of the host active materials utilized in SIB electrodes [63]. Palacín outlined these three charge insertion methods and their corresponding, concurrent structural alterations as displayed in Fig. 5 [10]. This example is nevertheless useful for comprehending insertion procedures in SIB materials even though it is based on host materials for LIBs. The reversible intercalation and extraction of Li^+/Na^+ in the lattice of materials with a layered structure is the primary mechanism of intercalation-type anode materials; however, the crystal and cation size have an impact [64]. Although the alloying reaction between lithium/sodium and the anode material is the mechanism of the alloying-type anode material, the volume change is more noticeable both before and after charging and discharging [65]. The size and lattice structure of the cations such as Mg, Na, Li, Zn, etc. are not especially necessary for the conversion-type anode materials, which involve a redox reaction between two materials as opposed to the conventional intercalation-type or alloying-type anode materials.

4. Cathode materials

The operational mechanisms of SIBs and LIBs are similar, which provides a solid basis for SIB research and development [66]. However, the distinctions between Na and Li present SIBs with a number of difficulties as well as new possibilities [67]. SIBs have lower voltages than LIBs because the sodiation potentials in the same materials are frequently lower than the lithiation potentials [68]. Furthermore, there is a widespread misconception that the Na ion should have slower diffusion kinetics than the Li ion due to its larger radius. An electrode that allows electric current to leave an electrical device, such a battery, is called a cathode. During the battery discharge in SIBs, the reduction reaction takes place at the cathode [69]. SIB cathode materials are essential for achieving exceptional performance, increasing SIB specific energy, and advancing their usefulness [70]. Polyanionic compounds, Prussian blue, layer transition metal oxides, and organic compounds are the main categories under which these cathode materials fall [71]. These cathode materials guarantee a reversible intercalation reaction that influences the cycle life since they show no structural change during intercalation [72]. However, because Na^+ ions have a larger size (coordination number 6: 1.02 Å) than Li^+ ions (coordination number 6: 0.76 Å), continual structural change is unavoidable during Na^+ ion intercalation into the host structure contact [73]. Additionally, even with brief contact to air, sodiated transition metal materials are very hygroscopic, so care must be taken to prevent hydration of the material, especially the surface, which leads to the creation of NaOH, which impairs electrode performance because of its insulating qualities [74]. Therefore, careful handling and moisture-free conditions are necessary for the manufacture of sodiated cathode materials and batteries.

4.1. Layered transition metal oxides

Transition metal oxides are becoming more and more popular as effective cathode materials for SIBs because of their strong electrochemical performance, affordability, and widespread availability [16]. Layered sodium transition metal oxides (LTMOs) Na_xMO_2 ($0.5 < x < 1$, M: Mn, Fe, Cr, Co, Ni, Cu, etc.) are inexpensive, low-toxic, have a remarkable theoretical specific capacity ($\sim 220\text{--}270 \text{ mAh g}^{-1}$), good ionic conductivity, and other advantages [75]. Layered Na_xMO_2 is one of the most popular SIB cathode materials because of these benefits. The sodium presence affects the structure of these transition metal oxides [76]. A three-dimensional tunnel structure is produced when the sodium content is 0.5 or less, while a layered structure is mostly formed when the sodium content is greater than 0.5. The crystal structures of layered oxide cathode materials used in LIBs and SIBs are similar, however there are minor differences since Li^+ and Na^+ have different ionic radii. Delmas et al. made the initial discovery of LTMOs in SIBs in 1980.

Depending on whether sodium ions occupy prismatic (P) or octahedral (O) configurations, LTMOs are primarily divided into two classes: P-type layered oxides and O-type layered oxides [77]. The P-type and O-type layered oxides are further classified into structural polymorphs of P2-, P3-, O2-, and O3-type (P2: ABBA type; P3: ABBCCA type; O2: ABAC type; and O3: ABCABC type) based on the atomic conditions of sodium and the number of different oxide layers stacking [78]. As shown in Fig. 6a, the numerical value represents the stacking arrangement of oxygen atoms within the structure, while the designations O and P indicate the locations of the alkali ions at the octahedral (O) or trigonal prismatic (P) sites [79]. TM ions are essential to the formation of these crystal formations [80]. To permit the intercalation and de-intercalation of alkali ions (Li^+ or Na^+), the TM framework should typically preserve its host structure [81]. However, a great deal of study has shown that during synthesis or electrochemical charging-discharging operations, TM ions frequently experience significant migratory problems [82]. In a lattice composed of hexagonal sheets, there are three potential locations for oxygen atoms, which are typically referred to as A, B, and C. The numbers "2" and "3" indicate how many layers make up a single cell unit. The crystal structure of P-type layered oxides is coupled in an edge-shared and face-shared fashion, while the sodium ions are accommodated on the prismatic (P) configuration. P-type layered oxides are further divided into P2-type and P3-type layered oxides based on the oxygen layer stacking order and the sodium-ion location of the surrounding environment [83]. On the other hand, sodium ions are accommodated on the octahedral (O) configuration in O-type layered oxides, where the crystal structure is joined in an edge-shared fashion [84]. Depending on the number of distinct oxide layers stacked and the sodium-ion environment of the surrounding area, O-type layered oxides are further divided into O2-type and O3-type layered oxides [85]. The better electrochemical performance of P2- and O3-type LTMOs has attracted a lot of interest lately. Larger triangular prismatic sites in the P2-type oxides provide broader transport channels and facilitate diffusion, enhancing rate capability [86]. In contrast, the larger sodium-ion content of O3-type oxides increases their specific capacities. In multi-layer, layered 3d transition metal oxides $\text{A}_x\text{TM}_2\text{O}_2$ ($\text{A} = \text{Li, Na}$; $\text{TM} = 3\text{d}$ transition metals; $0 < x \leq 2$), two TM ion migration paths are suggested: out-of-plane migration (Fig. 6b) and in-plane migration (Fig. 6c). Whereas the latter refers to the migration of TM ions from one TM site inside the TM layer to another, the former denotes the migration of TM ions from the TM layer to tetrahedral or octahedral sites in the AM layer. These migrations are closely linked to structural changes that have a major effect on layered oxides' electrochemical performance [87]. Since many O3 and P2 cathode materials have remarkable high capacities with roughly 40%–60% of the theoretical value ($\sim 120\text{--}150 \text{ mAh g}^{-1}$) and excellent cycle performance in the voltage range of around 1.5–4.1 V, P2-phase and O3-phase are often the ones that are explored the most [88]. Transition metal oxide materials have a number of disadvantages despite their benefits, such as poor material stability, air and water reactivity, and limited specific filling capacity [89]. Because of its tight framework structure, metal oxide can be used to explore high-density cathode materials. A transition metal layer is created by the edge-sharing connections between the transition metal components. Phase change increases polarization and modifies the energy barrier, which impacts sodium ion dispersion [90]. The stress caused by changes in the lattice constant causes the material to undergo substantial structural changes [91]. These structural alterations affect how the cathode active material interacts with the current collector, which eventually lowers the battery's cycle performance and energy efficiency [92].

Liang X. et al. created a specially designed reconstruction layer that combines gradient Ca^{2+} doping with a rapid ion conductor NaCaPO_4 coating to improve the layered cathodes' surface chemical and mechanical stability [93]. Consequently, the optimized cathode showed long-term cycling stability (81.0% capacity retention after 300 cycles at 0.5°C), remarkable rate performance (105.3 mAh g^{-1} at 10°C), and robust cyclability under challenging conditions such as low temperature

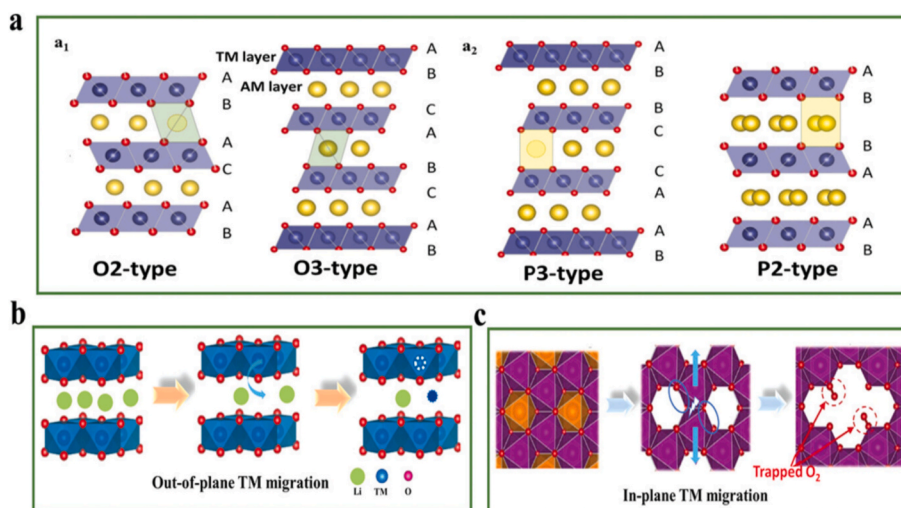


Fig. 6. Schematic representation of a) Layered oxide crystal formations of the O3, O2, P2, and P3 types. In multilayer oxide cathode materials, there are two types of TM ion migration: b) out-of-plane and c) in-plane. Reproduced with permission [79]. Copyright 2025, Wiley-VCH.

(−10 °C), high temperature (50 °C), and high voltage (4.5 V). Using interphase Ca²⁺ concentration gradient doping and NaCaPO₄ coating, they carefully developed an integrated technique to improve structural

stability (Fig. 7a). Fortunately, the chemical and mechanical stabilities of the layered oxides were improved by the addition of an interface reconstruction layer, greatly increasing each coated sample's cycle life

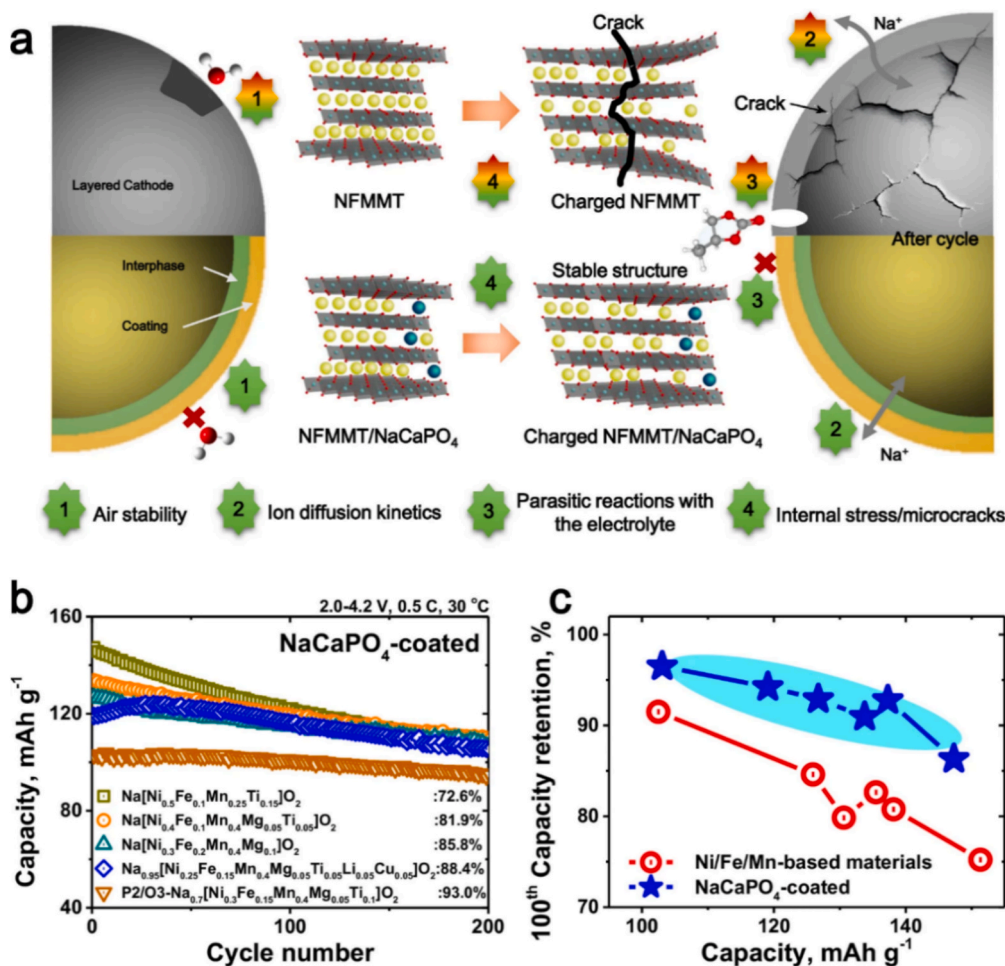


Fig. 7. A) Schematic illustrations of an integrated technique to improve structural stability using interphase Ca²⁺ concentration gradient doping and NaCaPO₄ coating. b) Other layered oxide materials with a surface reconstruction layer's cycle performance. c) Effect of interface reconstruction approach on Na storage performance of layered oxides. Reproduced with permission from the Creative Commons Attribution-Non Commercial-No Derivatives 4.0 International License [93]. Copyright © 2025, Nature.

(Fig. 7b). While there is still a negative association between cycling stability and attainable capacity, it is much smaller in the modified samples (Fig. 7c) and can be further diminished by optimizing composition and microstructure.

4.2. Polyanionic compounds

Polyanionic compounds are becoming more and more recognized as viable cathodic materials for SIBs due to their exceptional rate performance, high specific capacity, and great three-dimensional open structure stability [94]. The most common form of polyanionic compounds is $\text{Na}_x\text{M}_y(\text{X}_a\text{O}_b)_z\text{Z}_w$. X stands for elements like, Ge, As, S, Si, B, Mo, P, W, etc., Z stands for F, OH, etc., and M stands for metal ions having variable valence states, such as Ti, Mn, Fe, Nb, Co, V, Ni, Ca, Cr, Al, and Mg [95]. The polyhedrons of the polyanionic compounds are formed by joining the X and M polyhedrons via co-edges or co-points [96]. Na^+ ions are scattered across the system's gaps. These compounds are characterized by the presence of different polyanions integrated with transition metal frameworks, including pyrophosphate ($\text{P}_2\text{O}_7^{4-}$), sulphate (SO_4^{2-}), borate (B_5O_{10}), silicate (SiO_4), and phosphate (PO_4^{3-}) [97]. Because of their strong thermodynamic stability, polyanionic compounds exhibit excellent cycle performance and cathode safety. Another benefit is the slight structural alterations they undergo when sodium ions are inserted and removed because of their strong three-dimensional structure [98]. Nevertheless, the electrical conductivity and volume energy density of polyanionic compounds are modest [99]. The fast-charging capability and rate performance are negatively impacted by the low electronic conductivity [100]. The flow of ions and electrons in the electrodes as well as the stability of the electrodes play a major role in the growing demand for fast-charging SIBs [101]. A thorough study of polyanionic materials, including their crystal structures, fundamental physico-chemical characteristics, Na storage mechanism, and recent developments in polyanionic cathodes for SIBs, is crucial to solving these issues [102]. These issues can be optimized by ionic substitution or doping, coating, stoichiometric number variation in Na, size, and morphological design. Furthermore, certain X polyhedrons can raise the charge/discharge voltage by initiating the electrochemical activity of metals with varying valence states [100].

The $\text{k-NaVPO}_4\text{F}$ positive electrode material for sodium-ion batteries was created and produced by Shraer S.D. et al. Unlike favorite-type $\text{t-NaVPO}_4\text{F}$, which was previously known to be electrochemically inactive, $\text{k-NaVPO}_4\text{F}$ with the KTiPO_4 -type structure exhibits extended cycling stability, an attractive specific capacity of 136 mA h g^{-1} at 14.3 mA g^{-1} specific current, high C-rate performance due to low energy barriers and high diffusion coefficients. It is noteworthy that this material provides a practical specific capacity of 136 mA h g^{-1} at 14.3 mA g^{-1} , which is higher than the theoretical capacity of the commercialized $\text{Na}_3\text{V}_2(\text{PO}_4)_2\text{F}_3$ (128 mA h g^{-1}) and other known vanadium-based positive electrodes for one-electron $\text{V}^{3+} \leftrightarrow \text{V}^{4+}$ transition (Fig. 8), including NASICON-type $\text{Na}_3\text{V}_2(\text{PO}_4)_3$, $\text{Na}_3\text{V}_2(\text{PO}_4)_2\text{F}_3$ (or its closest relative, $\text{Na}_3\text{V}_2(\text{PO}_4)_2\text{O}_2\text{F}$, $\beta\text{-NaVP}_2\text{O}_7$, etc [103]. This material has one of the highest $\text{V}^{4+}/\text{V}^{3+}$ redox potentials for vanadium-based positive electrodes, at $4.3 \text{ V vs. Na}^+/\text{Na}$.

4.2.1. Phosphates

In general, polyanionic chemicals fall into a number of types, such as silicates, phosphates, pyrophosphates, sulfates, and mixed phosphates [104]. Because of the exceptional chemical and thermal stability of phosphate groups, phosphate systems have been the subject of the most research among these groups [105]. Following LiFePO_4 's successful commercialization, NaFePO_4 , a sodium counterpart, was investigated as a potential cathode material for SIB cathodes [106]. Triphylite (t-NaFePO_4) and maricite (m-NaFePO_4) are the two distinct phases of NaFePO_4 [107]. Moreover, t-NaFePO_4 contains a 1D Na^+ transport channel, an orthorhombic crystal structure, a Pnma space group, edge sharing with nearby $\text{FeO}_6\text{-PO}_4$, etc. (Fig. 9). The synthesis of t-NaFePO_4

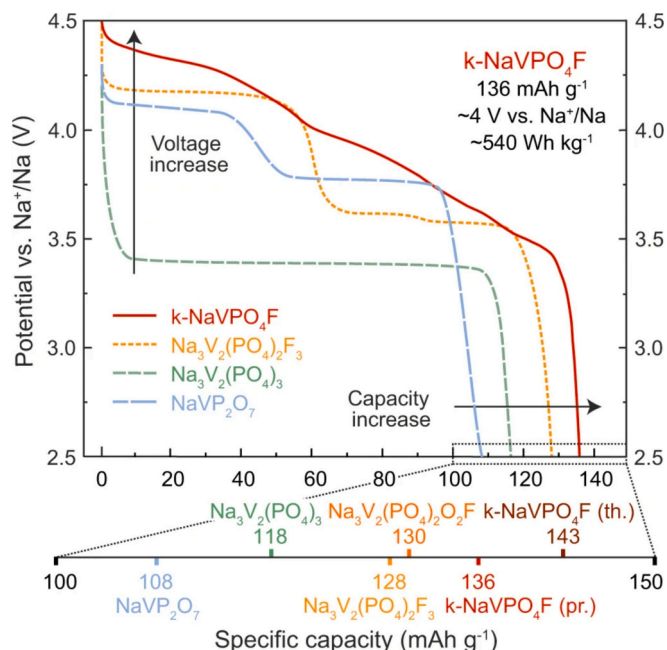


Fig. 8. Positive electrodes based on vanadium have voltage profiles. Reproduced with permission from the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License [103]. Copyright © 2022, Nature.

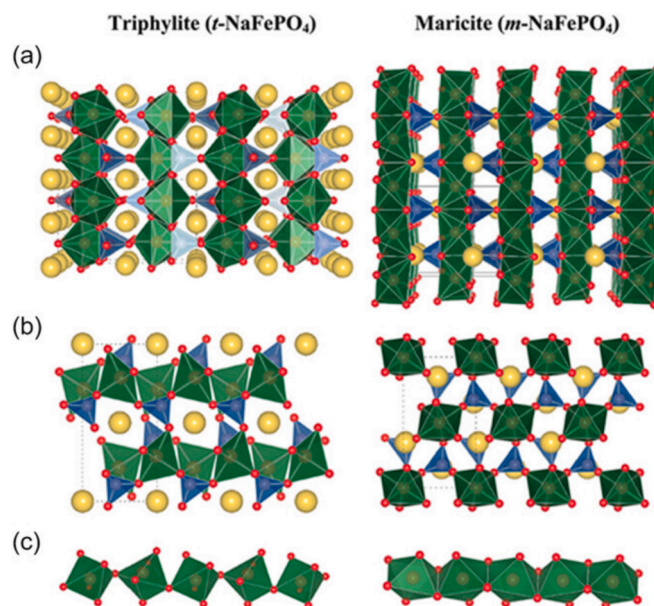


Fig. 9. A–C) structure of t-NaFePO_4 (left) and m-NaFePO_4 (right). Reproduced under the Creative Commons CC BY license, with permission from [4]. Copyright © 2023, John Wiley.

is possible below 400°C , while the synthesis of m-NaFePO_4 , which has edge-sharing $\text{FeO}_6\text{-FeO}_6$ units without a 1D t-NaFePO_4 , is possible above 500°C . t-NaFePO_4 is hence a better choice for a cathode material. Nonetheless, t-NaFePO_4 is a metastable variation of NaFePO_4 . It will change into the thermodynamically stable m-NaFePO_4 after undergoing an irreversible phase shift at temperatures exceeding 480°C [108]. Consequently, the conventional high-temperature solid-state technique is not appropriate for synthesizing t-NaFePO_4 . Rather, t-NaFePO_4 is usually produced through ion exchange [109]. In particular, iron-based phosphates have attracted significant attention for commercialization

because of their environmental friendliness and low cost. However, the relatively weak inductive effect of phosphate groups leads to a low operating voltage in phosphate cathodes, for example, O-NaFePO_4 (2.7 V vs Na^+/Na), $\text{Na}_4\text{Fe}_3(\text{PO}_4)_2\text{P}_2\text{O}_7$ (3.1 V vs Na^+/Na), $\text{Na}_3\text{Fe}_2(\text{PO}_4)_2\text{P}_2\text{O}_7$ (3.1 V vs Na^+/Na), $\text{Na}_2\text{FePO}_4\text{F}$ (3.0 V vs Na^+/Na), and $\text{Na}_3\text{Fe}_2(\text{PO}_4)_2\text{F}_3$ (3.1 V vs Na^+/Na) [110]. Due to the higher electronegativity of sulfur (2.6, Pauling Scale) compared to phosphorus (2.2), the inductive effect of SO_4^{2-} is stronger than that of PO_4^{3-} . This characteristic renders a significant advantage of sulfates over phosphates in terms of operating voltage. An example is $\text{Na}_2\text{Fe}_2(\text{SO}_4)_3$, which possesses a high working potential of up to 3.8 V (vs Na^+/Na) [111]. Moreover, in contrast to phosphates, which necessitate high-temperature calcination for synthesis, sulfates can be synthesized through various sustainable and cost-effective methods, such as low-temperature liquid phase precipitation reactions and solvothermal methods [112]. The convenient synthesis will significantly reduce the energy consumption and production cost, resulting in the economic and environmental benefits of sulfate cathodes for SIBs [113]. In the structure of sodium iron sulfate (NFS), Fe and Na serve as inexpensive and abundant redox-active centers and charge carriers, endowing SIBs with notable ecological friendliness and economic feasibility. Furthermore, its 3D framework provides open channels for ion/electron transport and effectively buffers the volume changes induced by sodiation/desodiation, thereby improving battery safety [114]. However, the material's intrinsically low conductivity results in sluggish reaction kinetics, which compromises the electrochemical reversibility and rate performance to some extent [115]. Worse still, NFS is sensitive to environmental humidity and prone to decompose at elevated temperatures due to the similar Lewis basicity of the oxygen atoms in SO_4^{2-} and water molecules [116]. Despite the serious challenges ahead, NFS cathode

materials still receive keen research interest and investment enthusiasm [117]. By optimizing synthetic method, tuning the morphology, and regulating microstructure, the structural stability and electrochemical performance of NFS can be significantly improved, thereby facilitating its development from laboratory research to mass production [118]. T. Gan et al., reported a novel continuous flow electrochemical solid-liquid coupling reactor-based sodium insertion technology to synthesize olivine-structured LiFePO_4 (LFP) [5]. The electrochemical performance of NFP at room temperature is shown in Fig. 9. Initially, NFP was cycled for four cycles at 0.1C within the electrochemical window of 1.5–4 V (vs Na^+/Na), followed by cycling at 0.2C for 100 cycles. At a discharge rate of 0.2C, the fifth cycle discharge capacity for NFP is recorded at 126 mAh g^{-1} , with a capacity retention of 86.5% after 100 cycles. The battery exhibits an initial Coulombic efficiency of 109%, and this efficiency remains consistently $\approx 100\%$ during subsequent cycles. The rate performance of the NFP battery is illustrated in Fig. 9b, where the battery underwent testing from 0.1C to 2C within a voltage range of 1.5–4 V (vs. Na^+/Na). As the discharge rate gradually increases, the recorded discharge capacities are $\approx 132, 123, 112, 104$, and 95 mAh g^{-1} , respectively.

For the initial two cycles, the charge-discharge curves of NFP clearly demonstrate two plateau regions at 2.94 and 3.11 V (vs Na^+/Na) corresponding to the solid-solution reaction ($2/3 < x < 1$, Na_xFePO_4) and the two-phase reactions ($0 < x < 2/3$, Na_xFePO_4), respectively (Fig. 10a). In the initial cycle of the synthesized NFP, an insufficient charge capacity was observed, notably pronounced in the charge capacity aligned with the solid-solution plateau ($2/3 < x < 1$, Na_xFePO_4). This further indicates that the sodium ion insertion in NFP falls short of reaching 100%, consequently resulting in an inadequate capacity for the solid-solution reaction plateau corresponding to Na_xFePO_4 within the

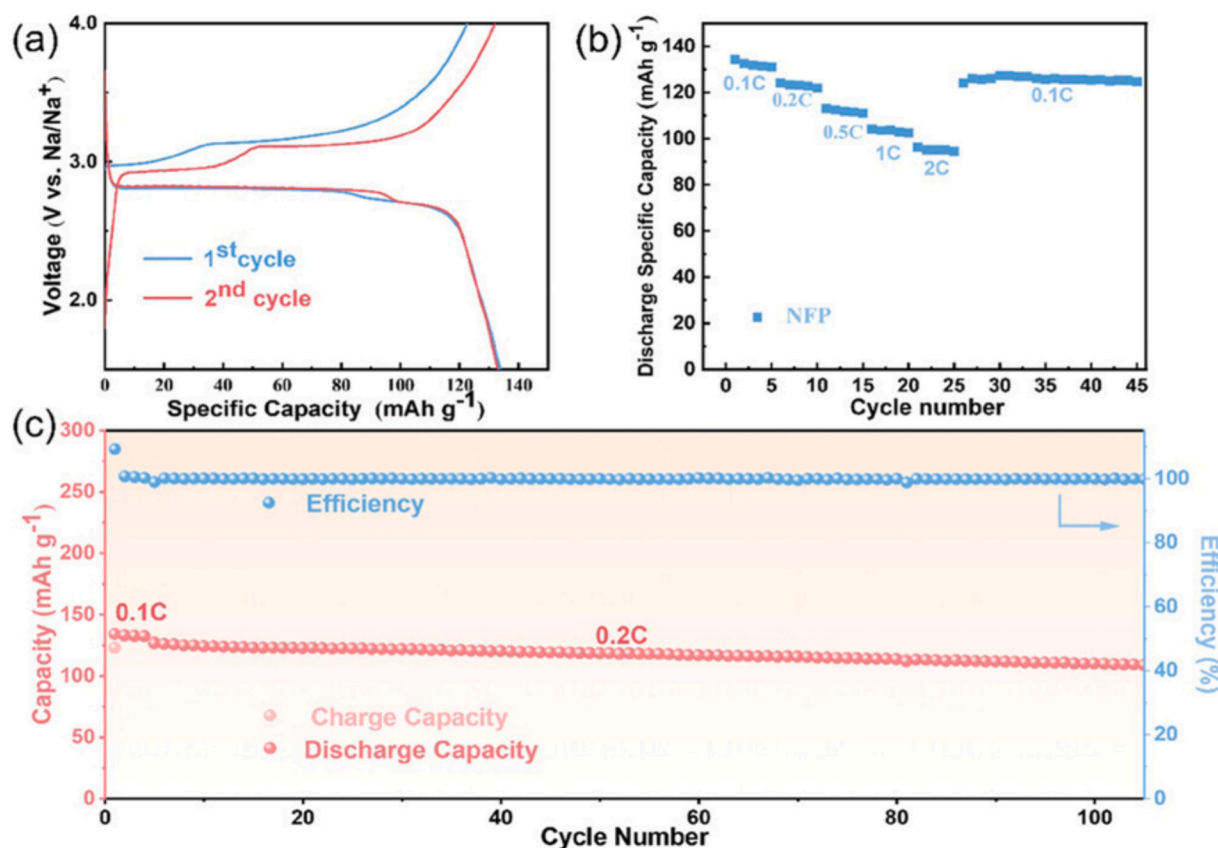


Fig. 10. Electrochemical performance of NFP half-cell. a) Charge-discharge curves of NFP for the first and second cycles at 0.1C, within a voltage range of 1.5–4 V. b) Rate performance of NFP within the voltage range of 1.5–4 V. c) Cyclic performance of NFP at 28 °C. Reproduced with permission [5]. Copyright 2024, John Wiley.

range of $2/3 < x < 1$ [119]. Hence, this approach reduces the environmental impact of chemical reagents and offers novel insights into the efficient, economical, and environmentally friendly recycling of LiFePO_4 and preparation of NaFePO_4 material.

4.2.2. Sulfates

Polyanionic compounds, layered transition metal oxides, and Prussian blue (white) analogues are examples of current mainstream SIB cathodes [120]. Due to their stable three-dimensional framework structures made of strong covalent connections, polyanionic compounds mainly phosphates, silicates, and sulfates show remarkable structural stability and the potential for a long lifespan [121]. These qualities make them excellent choices for creating long-lasting, highly safe energy storage batteries. Compared to most other mentioned cathodes, polyanionic sulfate cathodes have comparatively larger $M^{n+}/M^{(n-1)+}$ redox potentials due to the strong electronegativity of sulfur. Because of sulfates' hygroscopic qualities and lack thermal stability at high temperatures (i.e., above 350 °C), sulfate-based polyanions are not made using traditional high-temperature solid state procedures in aqueous media [122]. Using methods including ball milling, solvothermal synthesis, and a low-temperature solid-state approach, this has led to a low-cost, sustainable synthesis that lowers costs and energy consumption and helps commercialize it. Additionally, in sulfate polyanionic compounds $\text{Na}_2\text{M}(\text{SO}_4)_2 \cdot n\text{H}_2\text{O}$ ($M = \text{Fe, Mn, Co, Ni, etc.}$), the SO_4^{2-} group has poorer thermodynamic stability and a decomposition temperature below 400 °C [96]. For sulfate synthesis, the low-temperature solid-phase approach is frequently employed since SO_2 gas is readily emitted at temperatures higher than the breakdown temperature, leading to hazardous compounds and low chemical purity. Sulfate is strongly inducted because the S–O bonding is stronger than the P–O bonding, according to the Pauling electromagnetic principle. The redox potential of the material is so high due to the enhanced energy level cleavage that results from the hybridization of the d orbitals of transition metal ions with $\text{O}2p$ orbitals [123].

The theoretical specific capacity of sulfate materials is lower than that of phosphate because of the much lower charge-to-mass ratio of sulfate [124]. Sulphate materials can release their own special benefits in the realm of low-cost energy storage, even though they can barely be compared to marketed LiCoO_2 , LiFePO_4 , NCM-811, and so on in terms of practical applications. As a representative of this type of cathode among sodium iron sulfate (NFS), $\text{Na}_2\text{Fe}_2(\text{SO}_4)_3$ shows great promise for high-performance SIBs because of its many benefits, which include good temperature adaptability, high safety, excellent rate performance, long lifespan, and high voltage (3.8 V vs. Na^+/Na). As a result, NFS's low-temperature synthesis approach gives it a great overall performance and notable cost and energy efficiency advantages. It is emerging as a promising cathode to meet the growing demands of energy storage systems and electric automotive applications, especially in applications where high-rate capability, low cost, and high safety are crucial, like in stop-start batteries, when combined with the maturing industrial supply chain. However, NFS has a number of inherent problems, such as hygroscopic behavior brought on by thermodynamic instability and surface hydration reactions, intrinsically low electronic conductivity due to the electronically insulating nature of the polyanionic framework, and the frequent occurrence of impurity phases.

4.2.3. Silicates

Silicates, represented as Na_2MSiO_4 ($M = \text{Mn, Ni, Fe, Co}$), are attractive cathode candidates for large-scale energy storage due to the affordability and availability of Na, Fe, and Si raw materials on Earth [125]. Parasilicates, metasilicates, and orthosilicates make up the majority of the silicate family. Because of their low molecular weight, silicate compounds like lithium orthosilicate $\text{Li}_2\text{FeSiO}_4$ with a theoretical capacity of roughly 300 mAh g^{-1} generally have a greater theoretical capacity than other polyanions [126]. Mn was the first of the potential transition metals to be investigated in SIBs as $\text{Na}_2\text{MnSiO}_4$, which has a

suitably high-rate capability with a capacity of 125 mA h g^{-1} via the exchange of $1e^-$ at a rate of 0.1C (13.9 mA g^{-1}) between 2.0 to 4.0 V at 363 K. The complicated crystal structure and electrochemistry of orthosilicates make them difficult to use in practical applications. Moreover, analyze the structure of a silicate-based polyanionic compound using $\text{Na}_2\text{FeSiO}_4$ as an example [127]. While the sodium is hexacoordinated and forms a sublattice alone, the FeO_4 and SiO_4 tetrahedrons alternatively connect to form a solid framework in monoclinic $\text{Na}_2\text{FeSiO}_4$ materials (space group Pn); the Na ions are in a relatively disordered state in this structure. The structural framework's huge ion gap gives Na^+ more mobility, which raises the Na^+ diffusion coefficient. Therefore, even in the absence of the rapid ion transport channels, significant Na^+ diffusion in $\text{Na}_2\text{FeSiO}_4$ is still possible [128]. This also holds true for silicate cathodes based on Co and Mn. With a theoretical capacity of around 278 mA h g^{-1} , $\text{Na}_2\text{FeSiO}_4$ may reversibly insert and extract two Na^+ ions per unit [129]. Current research indicates that only the $1e^-$ reaction occurs because of electrolyte dissolution at potentials enough to insert/extract the second Na^+ , despite theoretical studies suggesting the feasibility of $2e^-$ reactions in silicate electrode materials. Because of their excellent physical and thermal stabilities, silicates were once widely used in the glass industry. Anti-site-exchange behavior was predicted for Co/Fe-compound sodium silicates, suggesting that they might make stable electrode materials [130]. The most promising silicate compound is $\text{Na}_2\text{FeSiO}_4$, which has a high theoretical capacity of about 276 mAh g^{-1} . Researchers have attempted to enhance its performance in a number of ways. Using a modified two-step process, M. Law et al. produced $\text{Na}_2\text{MnSiO}_4$ cathode material for sodium-ion batteries. The Na-hosting cathode's strain was lessened by the presence of fluorine with high electronegativity, and the volume change during the charge and discharge operation was only around 1.38%. $\text{Na}_2\text{MnSiO}_4$ was produced by Law et al. and shown a reversible capacity of about 100 mAh g^{-1} at 5C and a high capacity of about 210 mAh g^{-1} at 0.1C [2]. As seen in Fig. 11, the extra vinylene carbonate in the electrolyte enabled the creation of a passivated layer during cycling, which is what gave $\text{Na}_2\text{MnSiO}_4$ its exceptional performance. Consequently, the disadvantage of an unstable structure can be overcome by applying a surface passivation or protective layer. Compared to silicates based on lithium, sodium-based silicates are less studied. While sodium-based silicates lag behind due to their low electrical conductivity and impure phases formed during solid-state synthesis, lithium-based silicates are investigated with expected capacity due to a $2e^-$ -exchange mechanism. Sol-gel assisted solvothermal procedures and other solution-assisted synthesis methods, such as refluxing, have been used to create silicate-family nanoparticles in order to enhance the performance of sodium-based silicates.

4.2.4. NASICON

The NASICON (Sodium superionic conductor) structured materials $\text{Na}_x\text{M}_2(\text{PO}_4)_3$ ($M = \text{V, Ti; } x = 1, 2, 3$) are a class of materials well-known for their higher ion diffusion rates and stable 3D framework, which in turn provide good cyclic stability and satisfactory rate capability, in contrast to the olivine type of polyanionic compounds [131]. Polyanionic materials with a NASICON-type structure are particularly interesting as SIB cathodes because of their robust three-dimensional (3D) framework with large gap channels, which provide superior thermal and cycle stability than Prussian blue analogs or layered oxides. One of the first and most studied NASICON-type cathode materials for SIBs is $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ (NVP) [132]. It crystallizes in the trigonal system, such as R3c. In NASICON structures, sodium cations are positioned in interstitial spaces between corner-sharing PO_4 tetrahedra and VO_6 octahedra. All of the oxygen ions form PO_4 polyanions through strong covalent interactions with phosphorus, which give the 3D framework more stability than other cathode material types [133]. Furthermore, with a working voltage of 3.4 V versus Na^+/Na , which corresponds to the redox reaction of $\text{V}^{4+}/\text{V}^{3+}$, $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ provides a theoretical specific capacity of 117.6 mAh g^{-1} . Nevertheless, the biotoxicity and high cost of V element

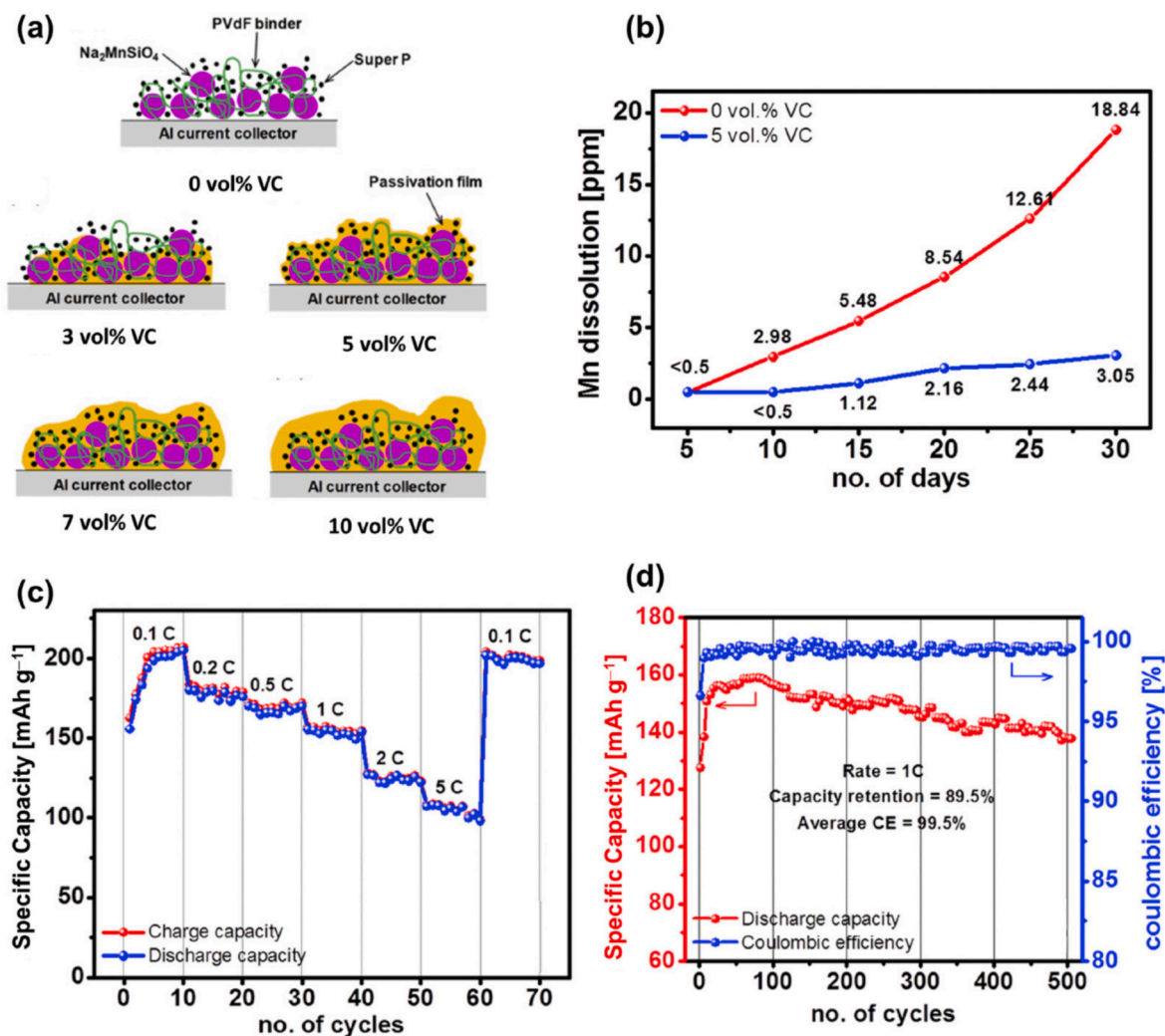


Fig. 11. (a) Illustration of the passivated layer on the Na₂MnSiO₄ cathode with additive vinylene carbonate (VC). (b) Mn dissolution of the Na₂MnSiO₄ electrode in electrolytes at room temperature. (c) Rate and (d) cycling performances of Na₂MnSiO₄ with 5 vol% VC at 1C. Reproduced with permission [2]. Copyright 2017, Elsevier.

in materials prevent its widespread use [134]. Therefore, the most promising way to lower the cost of polyanionic compounds is to lower the V content in NASICON-type materials or investigate alternative NASICON-type materials based on inexpensive transition metals [135]. Because of their affordability and ability to safeguard the environment, iron-based polyanionic materials have been considered the best cathode materials for SIBs [136]. Because of the inductive effect of the P₂O₇⁴⁻ group and the 3D robust skeleton with multiple Na⁺ migration paths, Na₃Fe₂(PO₄)₂(P₂O₇), a recently reported NASICON-type mixed polyanionic cathode material, has garnered increasing attention as a cathode for SIBs [137]. It has faster Na⁺ diffusion kinetics and a higher operating voltage than olivine-NaFePO₄. However, NFFPP's mean working voltage is only about 3.1 V versus Na⁺/Na, which is clearly lower than that of a cathode based on vanadium. Furthermore, polyanionic compounds' inherent low electrical conductivity prevents them from achieving superior electrochemical performance [138]. It is important to remember that Na₃V₂(PO₄)₃'s low intrinsic electronic conductivity, which leads to low specific capability, poor cycling stability, and poor rate performance, always limits its real use [139]. Numerous attempts have been made to enhance the electrochemical performance of Na₃V₂(PO₄)₃ by surface changes, including doping and carbon coating [140]. When employed as a cathode, carbon-coated Na₃V₂(PO₄)₃/C has proven to have outstanding performance and long-term stability. Hence, Na₃V₂(PO₄)₃-Na₃Fe₂(PO₄)₂(P₂O₇) (NVFPP/C/G) is a double-carbon-

layer decorated heterogeneous composite that was produced by J.-Z. Guo et al. as a cathode material for sodium-ion batteries [131]. The electrochemical de-intercalation/intercalation behavior of heterogeneous composite materials was tested using half-cells. The galvanostatic charge-discharge (GCD) curves of NVFPP-based materials decorated with different conductive carbons were compared in the potential window of 2–4.3 V vs Na⁺/Na at 0.1C, as shown in Fig. 12a. The two samples' cyclic voltammetry (CV) curves at a scan rate of 0.1 mV s⁻¹, and the redox peak's location matches the GCD curves comparison is shown at Fig. 12b. Naturally, the NVFPP/C/G cathode has the strongest redox peak. Furthermore, the rate and cycle performance of the NVFPP-based heterogeneous composite materials were evaluated and are shown in Fig. 12c–e. First, NVFPP/C/G materials have the highest specific capacity of 113.1 mA h g⁻¹ at 0.1C when comparing the rate performance of NVFPP-based materials at 0.1C and 15C. They also exhibit a high-rate capability of 70.4 mA h g⁻¹, indicating that the double-carbon layer can improve the material's electrical conductivity even at a high rate of 15C. The higher rate performance of NVFPP/C/G is also sufficiently demonstrated by the GCD curves at different speeds in Fig. 12d. At a high rate of 15C, the plateaus of the GCD curve may also be seen. Fig. 11e shows the cycling performance of the two samples at 5 C. NVFPP/C/G materials have a capacity retention rate of 78.6% after 500 cycles, which is slightly higher than NVFPP/C's 53.7%.

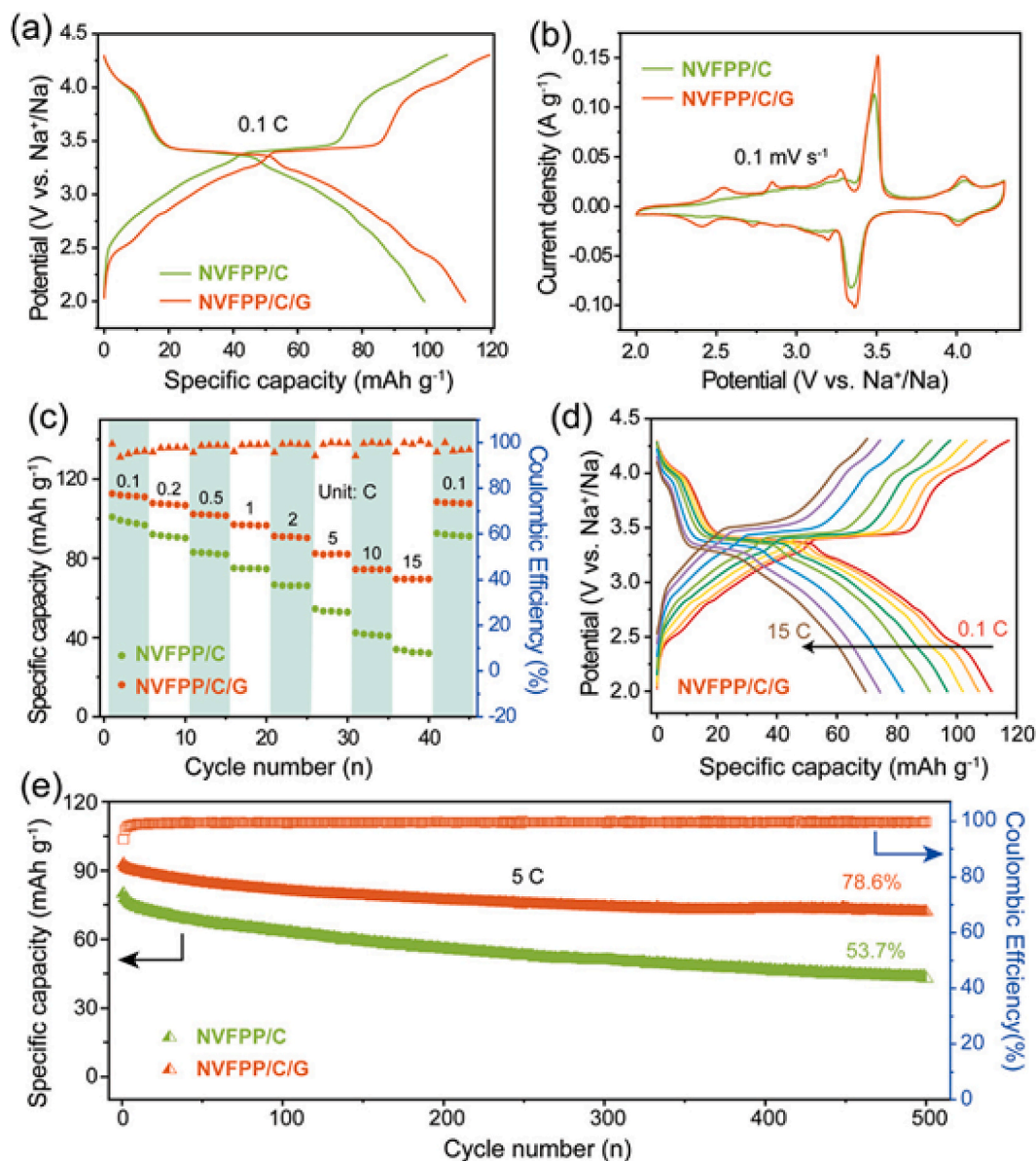


Fig. 12. NVFPP-based materials' electrochemical characteristics in the 2–4.3 V potential range vs Na⁺/Na. a) At 0.1C, the GCD curves; b) the CV curves at a scanning rate of 0.1 mV s⁻¹; c) rate performance from 0.1 to 15C; d) the GCD curves of NVFPP/C/G materials at various rates; e) cycling stability at 5 C after 500 cycles. Reproduced with permission [131]. Copyright 2022, John Wiley.

4.2.5. Pyrophosphates

Pyrophosphates are analogous to phosphates and generally derived from the decomposition of PO₄ under higher temperatures >500 °C and exhibit higher thermal stability as well as crystallinity [141]. Pyrophosphates have lower specific capacity due to higher molecular mass of P₂O₇ than PO₄, e.g., Na₂FeP₂O₇ has theoretical capacity of 97.2 mAh g⁻¹ while NaFePO₄ has theoretical capacity 154.2 mAh g⁻¹ [142]. The framework of pyrophosphates consists of MO₆ octahedra interconnected with P₂O₇ units providing Na⁺ migration channels. The pyrophosphates with dual sodium metals (Na₂MP₂O₇, M = Fe, Mn, V, Co, etc.) are extensively researched with corresponding crystal structures of triclinic, orthorhombic, tetragonal, and monoclinic phases [143]. Although the pyrophosphates of Fe, V, and Mn have better electrochemical activity than phosphates because to the increased stability of the P₂O₇⁴⁻ group, they have relatively low-rate capabilities, crystallinity, and theoretical specific capacities [144]. When exposed to air, orthophosphates break down between 500 and 550 °C to generate pyrophosphates, making pyrophosphates energetically more stable than orthophosphates at high

temperatures [145]. Since pyrophosphates always have larger unit cell characteristics than oxides, they have a larger spatial structure that permits free ion movement. An appealing material for energy storage, its open framework structure makes ion transport paths simple. The most prevalent kind of defect in this pyrophosphate framework is an antisite defect, in which the locations of the alkali cations and M ions alternate. This sort of defect requires a very low activation energy, indicating long-range and three-dimensional (Na ion) diffusion. In 2012, Yamada's group originally suggested Na₂FeP₂O₇ as a viable cathode material for sodium-ion batteries (SIBs) as a typical material in iron-based pyrophosphates. With channels for Na⁺ migration along the [011] direction, the space group of Na₂FeP₂O₇ is part of the trigonal P-1 phase and is made up of [FeO₆] octahedra and [PO₄] tetrahedra that share edges and corners to form a three-dimensional framework.

However, due to apparent voltage decline and comparatively low cycling stability, contemporary Fe-based pyrophosphates are still unsuitable for practical use [146]. It is possible that the structural rearrangement or the rearranging of the Na⁺ ions during the Na⁺

intercalation/deintercalation process is the cause of the typical dual voltage platforms of Fe-based pyrophosphate materials, where the platform near 3.0 V is made up of several tiny platforms. Keep in mind that for $\text{Na}_2\text{FeP}_2\text{O}_7$, there is a noticeable platform decay around 3.0 V, which ultimately causes the high platform capacity to decay as well as the total specific capacity to decline. To improve the electrochemical performance of $\text{Na}_2\text{FeP}_2\text{O}_7$, several tactics have been put forth recently, including surface modification, ion doping, defect introduction, and particle size manipulation. According to Pu X et al., a continuous solid solution, $\text{Na}_{4-\alpha}\text{Fe}_{2+\alpha/2}(\text{P}_2\text{O}_7)_2$ ($0 \leq \alpha \leq 1$), was discovered for the first time by mutual substitution of cations at center-symmetric Na3 and Na4

sites while maintaining the same crystal building blocks of anionic P_2O_7 [147]. Under galvanostatic charge and discharge conditions, the offsto-NFPO/C electrode could provide a notable discharge capacity of 83 mAh g^{-1} at a current density of 20 mA g^{-1} , as illustrated in Fig. 13A. Furthermore, the CV test in Fig. 13B revealed three peaks between 3 and 3.5 V and a symmetrical pair of peaks about 2.5 V, which represent a sequence of two-phase reactions and a single-phase reaction, respectively. The CV curves in Fig. 13C maintain their shape with a small shift in the redox peaks when the scan rate is continuously increased. The linear relationship between peak currents and the square root of the scanning rate ($v^{1/2}$) in Fig. 13D indicates that the Na-storage reaction in

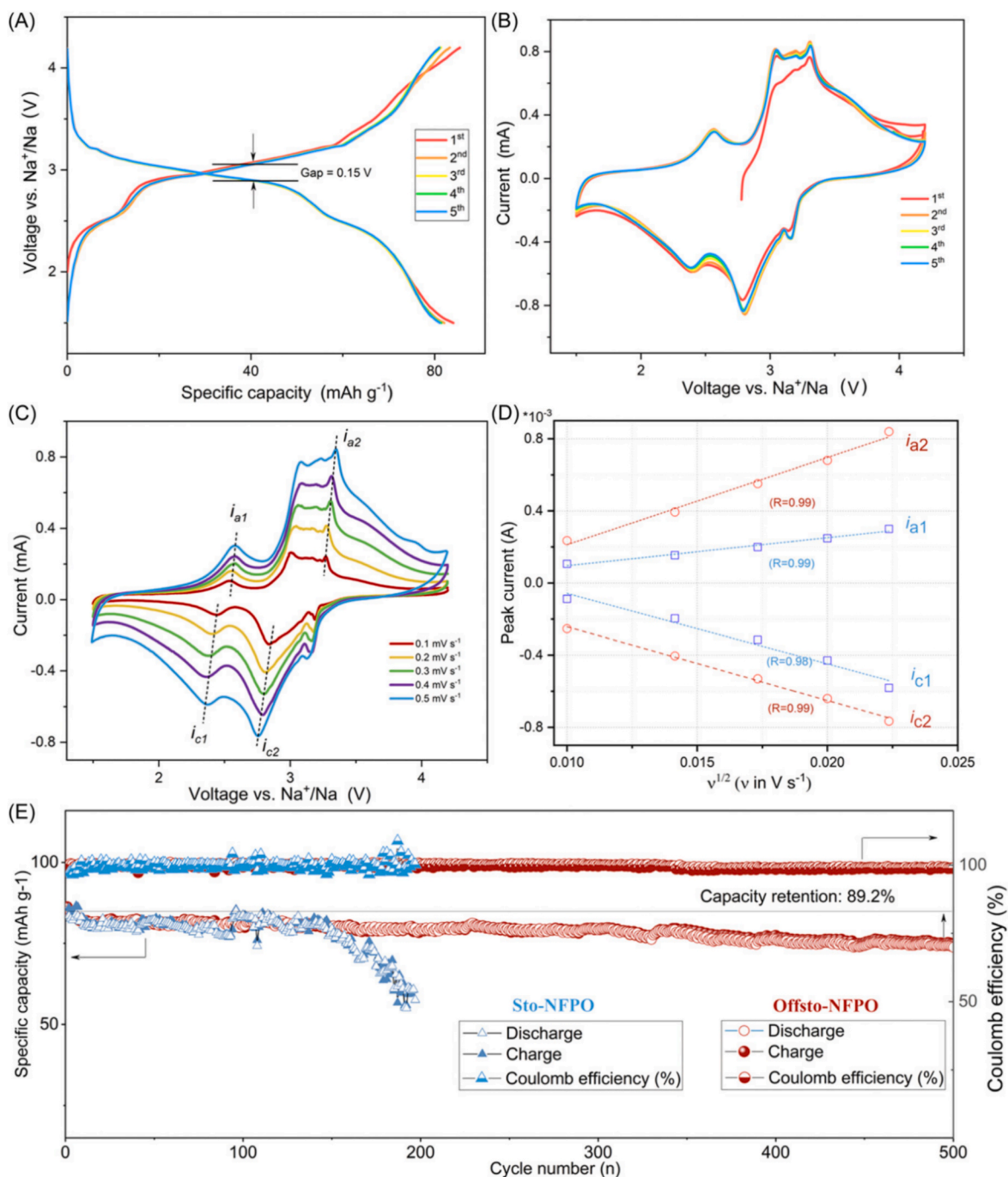


Fig. 13. Electrochemical reversibility in sodium-ion batteries. A) galvanostatic charge and discharge patterns at 20 mA g^{-1} current density, (B) C) CV curves at a higher scan rate from 0.1 to 0.5 mV s^{-1} , the first to fifth CV curves at a scan rate of 0.5 mV s^{-1} , E) Cycling performance over a long time at 20 mA g^{-1} for both offsto-NFPO and sto-NFPO, and D) The corresponding relationship of peak currents to the square root of scan rates (symbols reflect the measured data, while the dotted lines show the fitted values). Reproduced with permission under the terms of the Creative Commons Attribution license [147]. Copyright 2024, John Wiley.

offsto-NFPO is a diffusion-controlled process. As shown in Fig. 13E, sto-NFPO shows a rapid capacity reduction to 52 mAh g^{-1} even with fewer than 200 cycles. Conversely, when compared to the first cycle, the offsto-NFPO electrode exhibits remarkable retention of 89.2%, demonstrating the strengthened structure stability and intrinsic benefits of off-stoichiometric polyanionic frameworks when the solid solution range is expanded. A reversible capacity of 70 mAh g^{-1} is attained even after 500 cycles of an extended measuring procedure with a current density of 20 mA g^{-1} .

4.2.6. Orthophosphate

Orthophosphates are compounds that only contain one phosphorus unit, or the most basic form of phosphorus [148]. For SIBs, the general orthophosphate formula is NaMPO_4 , where M stands for transition metal. Inspired by olivine LiFePO_4 in LIBs system, its analogue NaFePO_4 is widely investigated for sodium storage in SIBs system. Differing from LiFePO_4 , the NaFePO_4 crystallizes into olivine and maricite phases. The crystal structures of both phases are orthorhombic with Pnma space group, but the arrangement of atoms is different from each other [149]. The framework of olivine phase is slightly distorted composed of FeO_6 octahedra along bc plane sharing edge/corner with PO_4 tetrahedra creating zigzag 1D path along b-axis for Na^+ diffusion and is electrochemically active. The maricite framework is composed of edge shared FeO_6 octahedra forming chains along b-axis connected to PO_4 tetrahedra by corner sharing forming 3D structure, so that there are no paths for Na^+ diffusion in the structure and is usually considered electrochemically inactive [150]. Compared to olivine, the maricite phase is thermodynamically stable, but because of its structural configuration, it lacks Na^+ ion transport channels, making it electrochemically inert [151]. The olivine phase features corner-shared octahedral units that are joined to tetrahedral units, while the maricite phase has neighboring octahedral units that share the edges and are further connected to tetrahedral units to share the corners. 125 mA h g^{-1} is the theoretical capacity of the olivine phase of NaFePO_4 , which is 80%.

4.2.7. Borates

Researchers have also paid some attention to borate polyanionic electrode materials because to their low molar mass, available resources, and environmental friendliness [152]. Borates have a brief history of use in rechargeable batteries, which may be attributed to research on LiFeBO_3 , the most basic electrochemical active borate. Following that, LiFeBO_3 and its variants are widely considered as promising cathode materials. Borates are made up of two basic structural units: the BO_3 triangular planar and the BO_4 tetrahedron. By sharing an O corner, edge, or plane, these two units can be joined to create different polyborate B_xO_y ions. $[\text{BO}_2]_\infty$, B_2O_4 , B_2O_5 , B_3O_6 , B_3O_7 , B_4O_{10} units, and others are well-known in addition to the fundamental BO_3 and BO_4 units. Borates exhibit two clear benefits when compared to other polyanion-type inorganic salts.

To begin, in contrast to silicates, phosphates, and sulphates, the B element in borates has two distinct coordination geometries with O atoms, whereas Si, P, and S are typically fourfold-coordinated with O atoms [153]. The number of prospective and available borates is therefore far greater than the other numbers. Second, LiFePO_4 , the most well-known and successful polyanion-type cathode material, has a theoretical capacity of only 170 mAh/g , which is less than the 200 mAh/g of the most basic borate-type cathode, LiFeBO_3 . Since borates have low stabilities, their synthesis is undoubtedly more challenging than that of the others. Furthermore, the simple BO_3 planar triangle and BO_4 tetrahedron can be used to create a variety of borate ions, which could make it challenging to prepare pure phases [154]. Borates' potential for energy storage has been less studied than that of other kinds of inorganic salts; the majority of the results that are currently accessible concern their use in LIBs. Accordingly, very few of them have anything to do with MIBs, SIBs, or zinc-air batteries. However, the majority of the borates under study were employed as cathode materials. Although not enough

research has been done on borate-type electrode materials, those that have been researched have shown encouraging electrochemical properties [155].

4.2.8. Mixed-Polyanion materials

The development of cathodes with a higher energy density for LIBs and SIBs has been the focus of intense research over the past 20 years. Because of the strong covalence nature of the M–O link, oxide cathodes have an inadequately high redox potential even if they can have a large reversible capacity [156]. Using polyanion cathode materials with a greater redox potential because of their sturdy framework and inductive action helps to overcome this drawback, resulting in longer cycle life and safe operation. Since less material is used per kilowatt-hour, increasing the cathode materials' energy density is an effective way to cut costs [156]. Notably, a useful tactic for investigating novel sodium storage structures with high energy density is the incorporation of different anions into the structure of polyanions to create mixed polyanion compounds (MPCs) [157]. Because of their longer lifespan, lower volumetric strain, and greater operating voltage than current polyanionic materials, mixed phosphates have recently attracted the attention of scientists as a new class of SIB cathode materials. The 3D channels in a mixed polyanion include four Na^+ sites, which enables rapid Na^+ diffusion in contrast to the 1D channels that are feasible in other polyanions. By adding various combinations of polyanions, the structure's tunability might potentially produce a unique cathode [158]. Their mutual compatibility can be used to create a range of hybrid polyanionic cathode materials with unique structures, including fluorinated phosphate, phosphate carbonate hybrids, and phosphate-pyrophosphate hybrids. The formula for the phosphate pyrophosphate hybrid polyanionic cathode material is $\text{Na}_4\text{M}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$, where M is Fe, Ni, Mn, Co, etc [159]. $\text{Na}_4\text{M}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ has a rhombohedral crystal structure with a space group of Pn21a. The MO_6 octahedra and PO_4 tetrahedra form a double chain through covertex connections, which is further connected to a laminar structure by pyrophosphate P–O–P bonding. Multiple electron redox activity and rich structural variation can result from these polyoxyanion combinations. Adjusting the ratio of various polyoxyanions can help adjust the degree of inductive action and, consequently, the redox potential [160]. Their ability to perform multiple electron redox reactions makes up for the limited capacity caused by the large molecular weights of polyanions (two electrons in $\text{Na}_3\text{MnPO}_4\text{CO}_3$, for example). $\text{Na}_4\text{Co}_3(\text{PO}_4)_2\text{P}_2\text{O}_7$, for example, may deliver high operating voltages above 4.5 V because of the inductive effect and right crystal structure [161]. Na^+ migration channels with very quick kinetics can be facilitated by their open framework (e.g., $\text{Na}_4\text{Fe}_3(\text{PO}_4)_2\text{P}_2\text{O}_7$). The resulting MPCs can display variable redox potentials in addition to extensive structural variety and multi-electron electrochemical processes. It is confirmed that the substantial inductive influence on the redox couples allows the injected anions to raise the redox potential values [162]. For example, the significant inductive influence of fluorine ions on the $\text{V}^{3+}/\text{V}^{4+}$ redox potential in fluorine-rich samples results in a greater $\text{V}^{3+}/\text{V}^{4+}$ redox potential of the $\text{Na}_3(\text{VPO}_4)_2\text{F}_3$ ($\sim 3.9 \text{ V}$) than the $\text{V}^{4+}/\text{V}^{5+}$ in $\text{Na}_3(\text{VOPO}_4)_2\text{F}$ ($\sim 3.77 \text{ V}$). Additionally, the $\text{V}^{3+}/\text{V}^{4+}$ redox reaction produces an exceptionally high working potential of about 4 V due to the increased inductive effect caused by the three $(\text{PO}_3)_3\text{N}$ groups surrounding every corner of the VO_6 octahedron in $\text{Na}_3\text{V}(\text{PO}_3)_3\text{N}$. The higher redox potential of the $\text{Na}_x\text{Fe}_y(\text{PO}_4)_3\text{-z}(\text{SO}_4)_z$ ($0 \leq z \leq 3$) solid solution compounds can also result from the inductive action induced by the strongly electronegative $(\text{SO}_4)^{2-}$ group. Furthermore, different sodium storage characteristics result from derivative compounds with modified valence bonds and crystalline structures. Consequently, it may be concluded that MPCs with appropriate architectures may have reasonable electrochemical performance. Zhao X. et al., developed the layered-columnar framework material $\text{NaFe}[\text{O}_3\text{PCH}(\text{OH})\text{CO}_2]$ using a one-step hydrothermal process in order to produce phosphoryl Na^+ -ion positive electrode material with a long cycle life and high-rate capacity [163]. Layered-columnar

structures are based on organic phosphoric acid, specifically its phosphate groups ($-\text{PO}_3$) that stretch outward along the organic chain and act as the stiff pillars of the structural framework (Fig. 14a). The synthesized $\text{NaFe}[\text{O}_3\text{PCH}(\text{OH})\text{CO}_2]$ sample is characterized by powder X-ray diffraction (XRD), and Rietveld refinement utilizing FullProf software is used to further evaluate the powder XRD patterns. The produced single-phase samples are of great purity, as seen in Fig. 14b, where the fitted curves have good reliability factor values ($R_{\text{wp}} = 9.56\%$, $R_p = 7.37\%$) and are well compatible with the original XRD patterns. Furthermore, Fig. 14c illustrates the refined structure of $\text{NaFe}[\text{O}_3\text{PCH}(\text{OH})\text{CO}_2]$, which is composed of sodium cations and polyanions and is generated by phosphoryl organic and Fe^{2+} ions. FTIR analysis is used to examine the chemical makeup of the produced $\text{NaFe}[\text{O}_3\text{PCH}(\text{OH})\text{CO}_2]$, as illustrated in Fig. 14d. More importantly, C-P and Fe-O-P bonds enable the $\text{NaFe}[\text{O}_3\text{PCH}(\text{OH})\text{CO}_2]$ with layered-columnar framework structure. It utilizes benefits such the stability of the inorganic structure, the flexibility of the organic structure, and electrochemical activity. It has a distinct layered structure with alternating organic and inorganic arrangements. Fig. 14e shows five peaks that correspond to P, C, O, Fe, and Na of $\text{NaFe}[\text{O}_3\text{PCH}(\text{OH})\text{CO}_2]$. According to the electrochemical

performance description, $\text{NaFe}[\text{O}_3\text{PCH}(\text{OH})\text{CO}_2]$ demonstrated a reversible specific capacity of 106.1 mAh g^{-1} , or 93.4% of the expected specific capacity, following full charge and discharge of sodium ions. Furthermore, the stable framework of the fully desodiated structure and the low strain of the structure during the charge and discharge allowed $\text{NaFe}[\text{O}_3\text{PCH}(\text{OH})\text{CO}_2]$ to have a long cycle life with satisfactory capacities of 61.6 mAh g^{-1} after the 1000th at current density of 240 mA g^{-1} . Furthermore, the redox of $\text{Fe}^{2+}/\text{Fe}^{3+}$ created a voltage plateau of 3.01 V in $\text{NaFe}[\text{O}_3\text{PCH}(\text{OH})\text{CO}_2]$, meeting the high voltage requirements for positive electrode materials. DFT simulations also show that layered-columnar structures containing the organic ligand of 3-phosphonopropionic acid have a lower sodium ion migration barrier. All-solid-state batteries are expected to employ these architectures. All things considered, this study demonstrates that producing phosphoryl layered-columnar materials is a practical way to produce high-performance positive electrode materials for sodium ion batteries.

4.3. Prussian blue analogues

For SIBs, Prussian blue analogues (PBAs) compounds have been

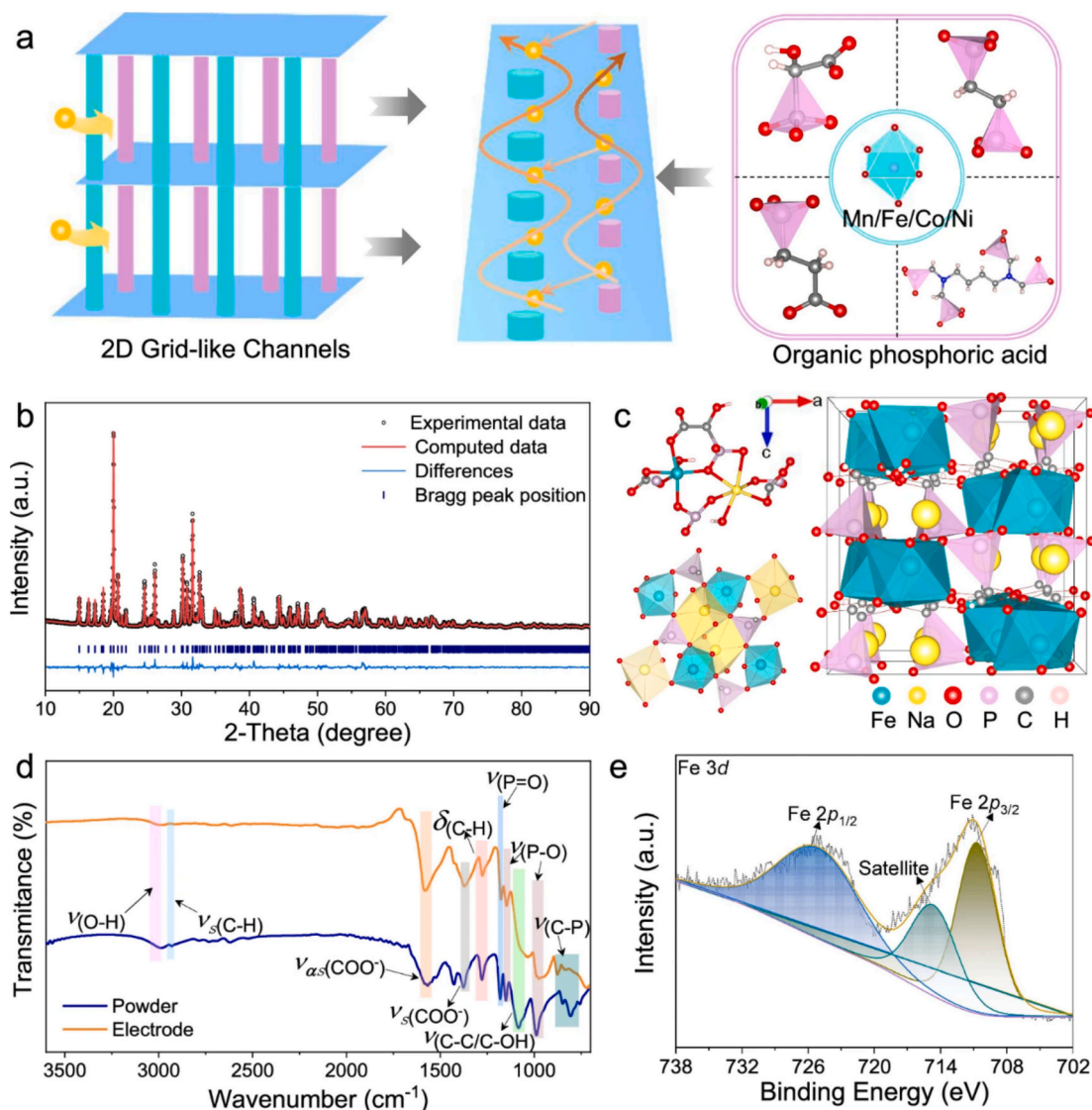


Fig. 14. A. the two-dimensional grid-like channels in the layered-columnar structure are made of organic phosphoric acid and metal. b. rietveld refinement of powder xrd cu $\text{K}\alpha$ ($\lambda_{\text{Cu-K}\alpha} = 1.54056 \text{ \AA}$) for the $\text{NaFe}[\text{O}_3\text{PCH}(\text{OH})\text{CO}_2]$ sample. c. $\text{NaFe}[\text{O}_3\text{PCH}(\text{OH})\text{CO}_2]$ crystal structure along (010) direction. d. FTIR of $\text{NaFe}[\text{O}_3\text{PCH}(\text{OH})\text{CO}_2]$ powder and $\text{NaFe}[\text{O}_3\text{PCH}(\text{OH})\text{CO}_2]$ electrode are compared. e. High resolution XPS OF Fe3d. Reproduced under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, with permission from [163]. Copyright © 2025, Nature.

acknowledged as efficient cathode materials [164]. They are very promising for this application because of their open and spacious structure, which makes it easier for Na ions to be accommodated and transported quickly. The common chemical formula for PBAs is $A_xM[R(CN)_6] \cdot zH_2O$, where A stands for mobile carrier ions (like Li^+ , Na^+ , K^+ , NH_4^+ , H^+ , Zn^{2+} , and Al^{3+}) and x is the stoichiometric coefficient, which usually ranges from 0 to 2, indicating the number of A sites occupied [164]. Transition metals (TMs) include both M (like Fe, Cu, Ni, Co, Mn, V, Zn) and R (like Fe, Cu, Ni, Co). The material's ability to store charge is improved by the valence changes of M and R during electrochemical charge and discharge cycles [165]. Hence, M ions produce $[MN_6]$ octahedra with nitrogen (N) atoms from cyanide ligands, which typically have high-spin electron configurations [164]. On the other hand, R ions form $[RC_6]$ octahedra with carbon (C) atoms, which are mainly in low-spin states. Because the N and C ligands have different electrical characteristics, there is a discrepancy in spin states. Compared to carbon, nitrogen is more electronegative and has a greater capacity to donate electrons, which strengthens the ligand field. The quantity of Na^+ ions and water concentration affect the crystal formations of PBAs, which can be cubic, monoclinic, rhombohedral, or trigonal. PBAs, which are primarily divided into two categories solid structures and hollow structures each of which contains a range of distinct morphologies, benefit greatly from the design of certain morphologies. The benefits of various morphologies will vary. Cubic arrangements, for instance, have more active cores. Particle sizes are more consistent in spherical formations [7]. Hollow structures are less likely to collapse because of their shorter ion diffusion and greater volume change adaptability (Fig. 15). Alkaline-rich PBAs often exhibit a monoclinic phase, whereas alkaline-deficient PBAs typically have a cubic structure. Following dehydration treatment, the phase structure will change to a trigonal or rhombohedral shape due to the decreased water content. The specific capabilities of PBAs as cathode materials for SIBs are determined by their chemical compositions (e.g., 85 mA h g^{-1} for a single-electron redox-active site (SE-PBAs M = Zn, Ni) and 170 mA h g^{-1} for double-electron redox-active sites (DE-PBAs, M = Mn, Fe, Co)). Given the high average discharge voltage (over 3.0 V vs. Na^+/Na), the theoretical energy density of DE-PBAs may exceed 510 Wh kg^{-1} , which is competitive with commercial $LiFePO_4$ utilized in LIBs [166]. The stability and rate capability of the battery are improved by the effective intercalation and deintercalation of sodium ions made possible by Prussian Blue's face-

centered cubic structure. Because high-temperature calcination is not necessary for the creation of PBA, manufacturing costs are lowered. PBAs are appropriate for large-scale manufacture and widespread application as reasonably priced cathode materials for SIBs due to their cost-effectiveness and structural advantages. PBAs are excellent sustainable cathode materials for SIBs and have several benefits. PBAs are well-suited for large-scale energy storage applications due to their affordability, the presence of numerous redox couples and highly reversible phase transitions, abundance of raw materials, their strong framework with three-dimensional diffusion channels ensures efficient alkali metal transport, simple synthesis methods and excellent cyclic stability further increases their efficiency in charge storage.

In general, defects, crystallinity, phase purity, and water content all have a significant impact on the PBA's electrochemical performance. The coordinated water in the lattice and the $Fe(CN)_6$ vacancy make it difficult for PBAs to achieve optimal electrochemical performance because they distort the crystal structure, accelerate the side reaction with the electrolyte, encourage the breakdown of organic electrolyte, and reduce ion/electron conductivity. Positively charged vacancies also reduce the specific capacity by lowering the absorption of sodium ions. PBAs include three different types of water molecules: interstitial and coordinated water, which are challenging to remove because of their physiochemical connection with the framework, and absorbed water, which is very easy to remove using a straightforward heating process below 120 °C. The absorption of water molecules can be decreased by prefilling the voids and increasing the salt concentration during synthesis. Numerous voids are created as a result of poor crystal growth caused by the widely utilized hydrothermal and coprecipitation production procedure of PBAs. A good way to create nanomicro-sized PBAs with fewer defects and increase the electrode's specificity is to fine-tune the fabrication process. The structural degradation brought on by the framework's defects and the dissolution of the transition metal ion, which can be somewhat mitigated by an appropriate surface coating, both impair the PBAs' cyclic stability. It is impossible to ignore safety concerns about cyanide release in very acidic environments and at high temperatures. Consideration is being given to properly treating chemical waste using extra oxidizing agent in order to completely oxidize cyanide to harmless compounds. The reversible specific capacity and energy density of batteries could be greatly increased by the open structural framework of sodium-rich PBA compounds. The presence of transition metals in the

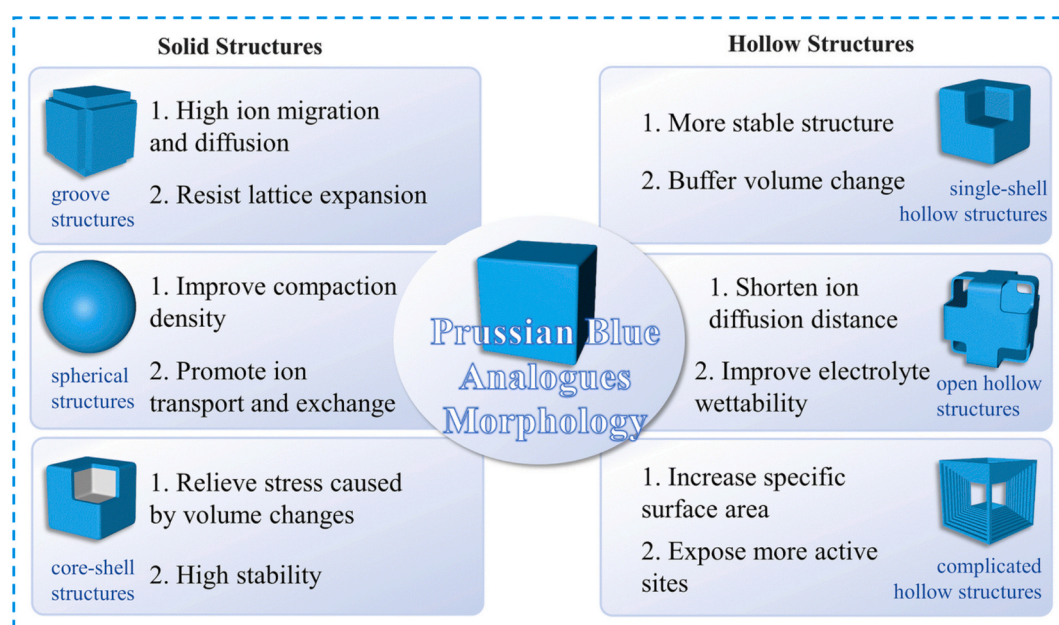


Fig. 15. Different morphologies of PBAs and their benefits. Reproduced under a CC BY-NC 3.0 license, with permission from [7]. Copyright © 2025, RSC.

PBA structure can enhance the electrochemical performance. Nonetheless, there are a number of issues with PBAs that must be resolved. Concerns about the environment, ease of use, cycle instability, transition metal ion dissolution, the difficulty of eliminating crystal water, and the requirement for effective ways to increase material capacity and coulombic efficiency are some of these. PBAs confront a number of obstacles in their implementation, despite their encouraging qualities. Low sodium storage content is the result of these, which include lattice vacancies and lattice water, which are caused by the co-precipitation reaction and irregular nucleation rates. It is challenging to activate the majority of Fe's low-spin states. Furthermore, during electrochemical cycling, irreversible phase transitions, lattice distortions, and Jahn-Teller processes can all lead to structural collapse in Mn-based PBAs, which can have a major effect on the electrochemical performance of PBAs. Collapse of the structural crystal can also result from parasite reactions at the electrolyte interface and stress during Na^+ insertion/desertion processes that is not immediately relieved. Pirayesh et al., developed a cathode material for SIBs by engineering a conformal, and nanoscale artificial cathode-electrolyte interphase (CEI) using a functionalized linked polymer (FLP). Hence, molecular layer deposition was used to deposit a homogeneous, molecule-level, thickness-controlled FLP layer over Prussian blue (PB), as shown in Fig. 16a. The bulk structure of the cathode material is not affected by the MLD process, as demonstrated in Fig. 16b, where the overall particle size and morphology did not change following the application of the FLP coating. A smooth, conformal artificial CEI layer with a thickness of about 8 nm covers the 25@FLP-PB particle uniformly, as shown in Fig. 16c. XPS was

also used to clarify the coating's surface chemistry. Two separate peaks can be seen in the 25@FLP-PB sample's high-resolution N 1 s spectrum (Fig. 16d). The clean and FLP-coated hexacyanoferrate samples' Fe K-edge X-ray Absorption Near-Edge Structure (XANES) spectra (Fig. 16e). These results provide a scalable path to high-performance, durable Na-ion cathodes by establishing fundamental design principles for artificial CEIs that emphasize conformality, controlled thickness and mechanical compliance.

The summary of recently developed different cathode materials for SIB applications are shown in Table 1.

5. Anode materials

The negative electrode is the one with the lowest potential. In SIBs with solvated Na^+ as the major charge carrier, the reduction potential of sodium (-2.71 V versus SHE) gives the thermodynamically lowest electrode potential. To increase the cell's energy density, it is therefore optimal to have a negative electrode with the highest specific capacity and, in theory, the lowest potential [190]. The most suitable negative material from a thermodynamic standpoint is metallic sodium, which has the lowest cell potential and a large theoretical capacity of 1165 mAh g^{-1} . The use of metallic electrodes in room temperature secondary batteries has not been accomplished to a significant degree due to safety issues [191]. This is due to the formation of a solid electrolyte interphase (SEI) or unstable surface passivation layer on the electrodeposited sodium surface. Thus, cycleability is limited by the ongoing reactivity of freshly created surfaces with the electrolyte during plating. Dendritic

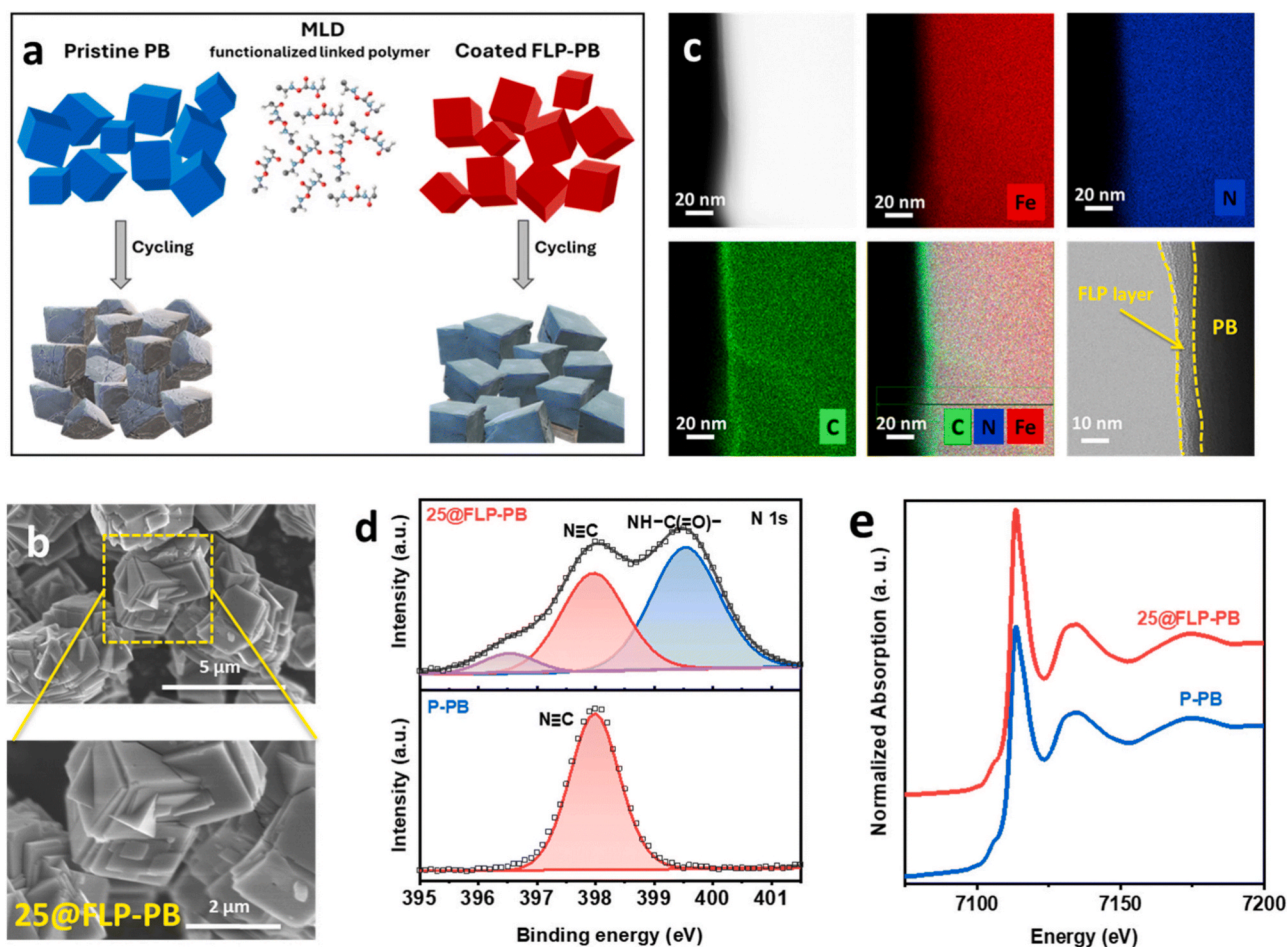


Fig. 16. (a) Diagram of the MLD procedure for depositing an FLP film to PB cathodes. (b) 25@FLP-PB particle SEM image. (c) TEM image of a 25@FLP-PB particle at high resolution. (d) The 25@FLP-PB sample's N 1 s XPS spectrum. (e) Fe K-edge XANES spectra of coated and pristine PB samples. Reproduced under a CC BY-NC 3.0 license, with permission from [167]. Copyright 2026, RSC.

Table 1
Summary of recently developed cathode material for SIBs.

Cathode material	Cyclability	Capacity (mAh g ⁻¹)	Voltage (V)	Refs.
P'-Na _{0.67} [Ni _{0.1} Fe _{0.1} Mn _{0.8}]O ₂	80%/200 cycles/3C	220	1.5–4.3	[168]
Na ₃ (VOPO ₄) ₂ F	91.3%/300 cycles/1C	110	2.4–4.4	[169]
Na _{1.93} Mn[Fe(CN) ₆]•2.38H ₂ O (NMF-H)	74%/500 cycles/1C	151.6	2.0–4.2	[170]
Na _{0.44} Mn _{0.94} Li _{0.06} O ₂	83.6%/300 cycles/1C	106.1	2.0–4.0	[70]
MnHCF@Fe/MnHCF	95.5%/400 cycles/1 A g ⁻¹	134.2	2.0–4.0	[171]
Na ₂ Fe(SO ₄) ₂ @rGO/C	81.5%/200 cycles/0.5C	85	2.0–4.5	[172]
Ni-doped Na _{0.44} MnO ₂	77.3%/100 cycles/0.3C	85.5	3.0–4.0	[173]
Na _{3.32} Fe _{2.34} (P ₂ O ₇) ₂ /C	92.3%/300 cycle/0.5C	100	1.5–4.0	[174]
Na _{0.44} Mn _{0.97} Mo _{0.03} O ₂	77.8%/100 cycles/1C	178.9	2.0–4.0	[175]
FeMnCu@PANI	~90%/500 cycles/10C	107	2.0–4.0	[176]
Na _{1.56} Fe[Fe(CN) ₆] _{0.90} □ _{0.10} •2.42H ₂ O	69.8%/200 cycles/0.5C	71.9	2.0–4.0	[177]
Na _{0.72} Li _{0.24} Mn _{0.76} O ₂	40.7%/80 cycles/0.05C	231	1.5–4.5	[178]
Na ₃ V ₂ (PO ₄) ₃ /rGO	>80%/8000 cycles/3 A g ⁻¹	87	2.5–4.0	[179]
Na _{0.44} Mn _{0.6} Ni _{0.3} Cu _{0.1} O ₂	80%/500 cycles/0.1C	149	1.5–4.0	[180]
PB@GO-PVP(MR)	~72.7%/500 cycles/0.2C	165.2	2.0–4.2	[181]
Na _{0.44} Mn _{0.95} Mg _{0.05} O ₂	70%/800 cycles/2C	105	2.0–3.8	[182]
Cs-doped KMnHCF	90.1%/200 cycles/2C	100.1	2.0–4.0	[183]
Ni ₃ (BO ₃) ₂	85.4%/50 cycles	304.4	0.1–3.0	[184]
Na ₂ Ti ₆ O ₁₃ -coated Na _{0.6} MnO ₂	86.7%/100 cycles/2C	175.6	2.0–4.1	[185]
MnCoNiFePB	56.7%/100 cycles/1C	75.4	2.5–4.5	[186]
Li ₂ FeSiO ₄	~40%/10 cycles/0.05C	126	1.0–4.1	[129]
CuPB/FePB	92.5%/800 cycles/1C	122.57	2.00–3.75	[187]
Na _{0.8} Mn _{0.55} Ni _{0.25} Fe _{0.1} Ti _{0.1} O ₂	80.2%/100 cycles/0.1C	154.6	1.5–4.3	[188]
Na ₄ Fe ₃ (PO ₄) ₂ P ₂ O ₇ @MCNTs	95%/1200 cycles/2C	115.7	1.5–4.0	[117]
Na ₃ V _{1.8} (CrMnFeZnAl) _{0.2} (PO ₄) ₃	80%/3000 cycle/10C	119.8	2.5–4.3	[189]

development during (dis)charge cycles presents additional difficulties. These result in heat runaway and internal short circuits if left unchecked. It is significant to highlight that the use of metallic anodes in rechargeable room temperature lithium-based or sodium-based battery systems has been an unresolved issue for nearly fifty years [192]. Finding an appropriate electrolyte or protective surface layers that hold steady at these extremely low potentials is the hope. After metallic Na struck a brick wall, the quest for a substitute anode shifted to active material hosts that could inject charges at potentials just above 0 V in

contrast to Na⁺/Na. By doing this, safety issues brought on by metallic plating are avoided. For instance, the electrolytic breakdown of ethylene carbonate in LIBs results in a very advantageous SEI that is electrically nonconductive and stops more electrolyte breakdown [193]. The unavailability of a relatively inexpensive, low potential anodic material for SIB applications and the discovery of graphite as a strong, high specific capacity anode material for LIB were the main obstacles to the advancement of SIB research [194]. The anode material undergoes a reduction reaction during charging and an oxidation reaction during discharge, usually with a lower reduction potential. An anode material should have high reversible capacity, safety, stable structure, low cost, low reduction potential and high ionic conductivity [195]. Currently, there are five primary categories of anode materials for SIBs such as alloying, intercalation, organic, conversion and MXene materials. The selection requirements for anode materials of sodium ion batteries are as follows: There should be a high potential difference between the anode material and the cathode material, which can ensure a higher potential output when the positive and negative electrode materials are matched. Therefore, the lower the redox potential of Na⁺ in the anode material, the better, and the closer the potential to Na, the better [196]. The anode material should also have a high ion diffusion rate to ensure a high specific capacity. The anode material should have certain structural stability to ensure the cycling stability of the battery during the charging and discharging process. The anode material should be selected with high conductivity, which can support the high current charging and discharging requirements of the equipment and minimize the polarization of the battery. Anode materials must have a strong, well-structured surface in order to support a stable solid electrolyte interface layer, which is necessary for preserving battery cycle life. Additionally, the anode material needs to have chemical stability to prevent irreversible reactions with the electrolyte and reduce the battery capacity. The anode material must be environmentally friendly, harmless to the human body, and the price should be as low as possible.

Sodium storage for anode materials typically involves three types of mechanisms: alloying/de-alloying events, conversion processes, and intercalation/de-intercalation reactions. It is commonly recognized that anode materials with high energy densities, such as metals and their alloys, phosphorus and phosphides, oxides, and sulfides/selenides, can store Na⁺ through an alloying or conversion reaction. However, these anode materials often have low cycling life due to their significant volume expansion, which results in inferior electrochemical performance and has limited the practical applications of sodium ion batteries up to this point. In terms of safety, intercalation-based anode materials have clear advantages over alloying or conversion-based materials, including natural abundance, long-term cycling stability, and environmental friendliness. Generally, in this section the recent developments in anode materials for SIBs including alloy-based, intercalation-based, conversion-type are presented. Moreover, organic, and MXene materials are discussed separately.

5.1. Intercalation-Based materials

SIBs anode material, which typically has a low reduction potential, is crucial to the loading and release of sodium ions; oxidation reactions take place during discharge and reduction reactions happen during charging [197]. Power density, rate performance, and other aspects of the battery's overall dynamic performance are directly impacted by the anode material's efficiency. In terms of safety, intercalation-based anode materials have clear advantages over alloying or conversion-based materials, including natural abundance, long-term cycling stability, and environmental friendliness [198]. The cyclic uptake and release of sodium ions within layered host structures is at the heart of the intercalation chemistry of SIB anodes, which provide a more affordable and abundant substitute for lithium-based systems. In contrast to lithium, sodium has a bigger ionic radius and different solvation behavior, which makes it difficult to use traditional graphite hosts but allows for special

co-intercalation pathways [199]. A thermodynamic barrier prevents bare sodium ions from being inserted into graphite, however solvated sodium species can co-intercalate within graphite galleries with great reversibility and quick kinetics. This mechanism causes the interlayer gap to expand and contract noticeably, demanding the use of specialized binder formulations and electrode configurations to control volume differences. According to a 2000 study by Stevens and Dahn, hard carbon (HC), one of the common intercalation-type anode materials for SIBs, has a greater interlayer spacing of 0.352 nm than normal graphite, which has an interlayer spacing of only 0.335 nm [200]. This could help Na^+ storage by achieving a high reversible capacity of about 300 mAh g^{-1} , which is comparable to graphite's lithium storage capacity. Over the past 20 years, titanium-based compounds including TiO_2 , $\text{Li}_4\text{Ti}_5\text{O}_{12}$, $\text{Na}_2\text{Ti}_3\text{O}_7$, and $\text{NaTi}_2(\text{PO}_4)_3$ have drawn a lot of attention in addition to carbonaceous materials because of their excellent chemical stability, nontoxicity, ease of synthesis, and high natural abundance (Fig. 18). These titanium-based materials have been widely used in recent years as anode materials in SIBs because they can reversibly store Na^+ through intercalation/de-intercalation processes with stable cycling lifespans, moderate energy density, and appropriate operating plateaus. TiO_2 has been used in many different technical fields, such as environmental cleanup, energy storage systems, nanotherapeutics, and dye-sensitized solar cells because of its special qualities, which include nontoxicity, natural abundance, affordability, and remarkable stability [201]. TiO_2 has been considered one of the most promising anode materials in the field of energy storage systems in addition to the previously mentioned advantages because of its mild sodium insertion potential at roughly 0.6 V (vs. Na^+/Na), which can resolve the problem of sodium dendrite formation. As is well known, the electrochemical performance of SIBs is significantly influenced by the phase structure of electrodes. In addition to the aforementioned benefits, TiO_2 has been regarded as one of the most promising anode materials in the field of energy storage systems due to the mild sodium insertion potential at about 0.6 V (vs. Na^+/Na), which can solve the issue of sodium dendrite formation [202]. As is generally known, the phase structure of electrodes has a major impact on the electrochemical performance of SIBs.

Bronze, rutile, baddeleyite, anatase, TiO_2 -ramsdellite, columbite, brookite and hollandite are polymorphs of TiO_2 . However, amorphous

TiO_2 , TiO_2 -H (hollandite), TiO_2 -B (bronze), rutile and anatase shows electrochemical activity when used as anode materials for SIBs [61]. The most well-known structures of TiO_2 are the naturally occurring brookite, anatase and rutile polymorphs shown in (Fig. 17), while numerous other artificially produced forms have been reported. Every polymorph has unique physical characteristics. Anatase and brookite are metastable at all temperatures and will change into rutile when heated, however rutile is the most stable phase of the three polymorphs in ambient settings. According to certain studies, the stability of different TiO_2 phases depended on the size of the particles. For nanoparticles smaller than 11 nm, the most thermodynamically stable phase is anatase. For nanoparticles larger than 35 nm, rutile has been determined to be the most stable phase; for those between 11 and 35 nm in size, brookite is the most stable phase.

Since the TiO_2 phase is the most important factor influencing the materials' characteristics, numerous studies have tried to comprehend the phase transition between anatase, rutile, and brookite. Anatase is typically the first crystalline TiO_2 phase to develop; calcination yields rutile. Due to the breaking and reformation of bonds, the transition from the anatase to the rutile phase is both reconstructive and irreversible. According to reports, when anatase is calcined in air, it turns into rutile at about 600°C . However, according to certain study, the transformation temperature might vary significantly between 400 and 1200°C depending on the raw materials, synthesis techniques, and heat flow conditions. When heated to around 800°C , brookite changed into the rutile phase. Rutile is the most thermally stable of the three polymorphs, having a melting point between 1830 and 1850°C . When the sample is heated above 1000°C for an extended length of time, TiO_2 is always tough to dissolve or react. Because phase transformation can change the catalyst's characteristics and performance, it is quite interesting.

In commercial lithium-ion batteries, graphite is the typical anode material. Graphite has a theoretical capacity of 372 mA h g^{-1} for storing lithium. In lithium-ion batteries, graphite materials have exceptional electrochemical characteristics, including a flat potential platform, a relatively low reversible capacity, an outstanding voltage window, good cycling behavior, high coulomb efficiency, and more [204]. However, graphite cannot be directly used in the development of anode materials for sodium-ion batteries because its structure prevents sodium ions from moving freely within it. Other carbon-based materials with adequate electrochemical capacity and recyclability are being sought for in addition to graphite for usage as negative electrode materials in sodium-ion batteries [205]. The carbon-based anode materials used in SIBs are primarily sp^2 hybrid, which exhibits distinct physical, chemical, and electrochemical properties due to variations in crystallinity and carbon layer arrangement. The term "carbon-based materials" mostly refers to amorphous carbon materials, which fall into two categories based on how difficult it is to graphitize: soft carbon and hard carbon [206]. The soft carbon material, which can be utilized as an anode material for SIBs, is a carbon material that is structurally in between graphite and hard carbon. It is often made up of amorphous or turbo-layered carbon. Na^+ intercalation can result in the irreversible growth of local and macroscopic structures, and soft carbon is less effective than hard carbon [207]. Soft carbon is used as a raw material for the production of artificial graphite or as a coating material to convert to natural graphite, alloy, and other anode materials; it is not directly utilized as an anode material [208]. Soft carbon, which often refers to amorphous carbon that may be graphitized following high temperature treatment (over 2800°C), low crystallinity, tiny grain size, large crystal surface spacing, and good electrolyte compatibility, is extremely easy to graphitize [209]. Hard carbon, also known as tough to graphitize carbon, is typically defined as carbon that is challenging to fully graphitize even after high temperature treatment (over 2800°C) and whose disordered structure is challenging to remove at high temperatures [210]. Soft and hard carbon have no noticeable structural boundaries when treated at low temperatures (1000 – 1600°C), and they can be collectively referred

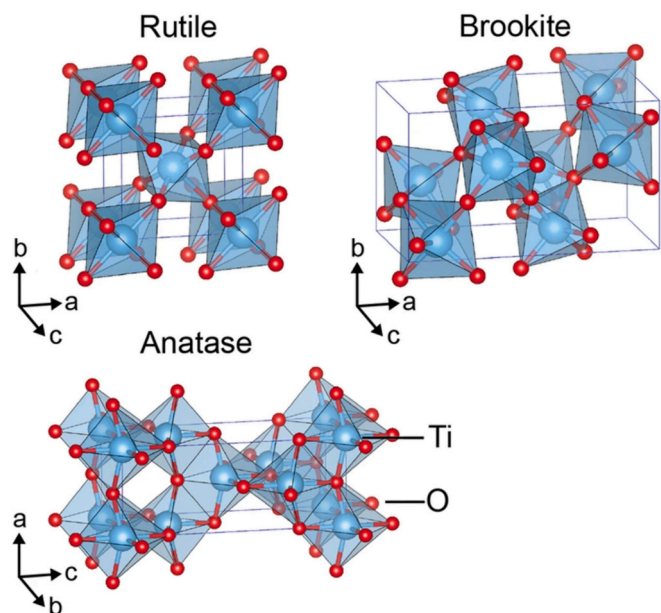


Fig. 17. TiO_2 polymorphs with crystal structures: anatase (tetragonal, $I4_1/amd$), brookite (orthorhombic, $Pbca$) and rutile (tetragonal, $P4_2/mmm$). Reproduced under the terms of a Creative Commons Attribution 4.0 International License [203]. Copyright 2024, Nature.

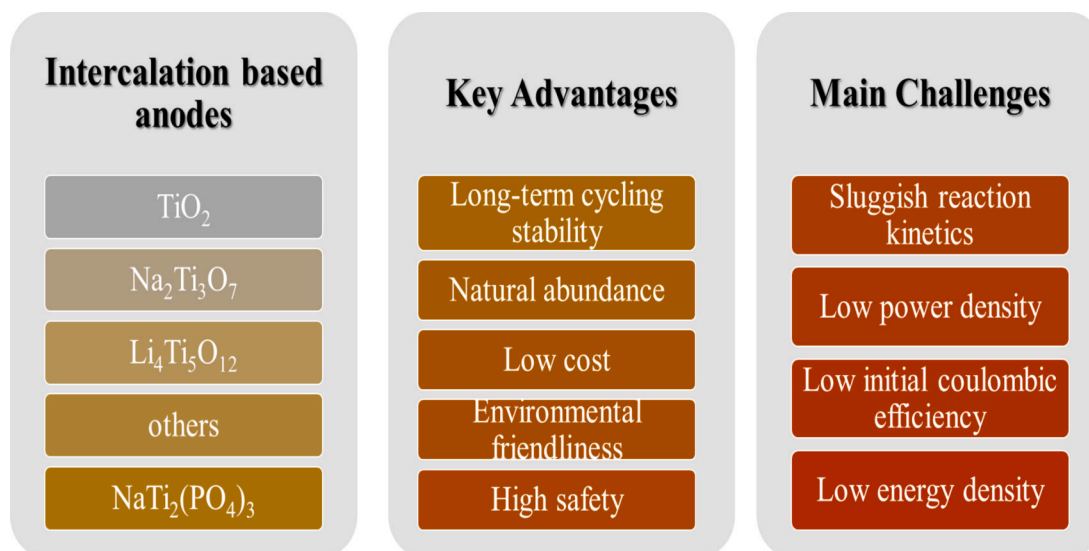


Fig. 18. Schematic representation of intercalation-based anodes in terms of types, merits, and challenges.

to as amorphous carbon. Soft carbon has a greater irreversible capacity and superior electrical conductivity. Soft carbon is inexpensive, has an abundance of raw materials, good electrical conductivity, and a high degree of structural regularity. The lower sodium storage capacity of soft carbon is approximately 220 mA h g^{-1} . However, the irreversible capacity of the first charge and discharge is high because sodium ions tend to alter the layer spacing during the reaction [211]. Additionally, the material's salt storage capacity will be diminished, the layer spacing will be decreased, and the soft carbon is easily graphitized at high temperatures. Hard carbon is more expensive than soft carbon, but it has a higher specific capacity and first charge and discharge efficiency. A significant portion of HC's remarkable specific capacity nearly 300 mAh g^{-1} , which is comparable to graphite in LIB occurs at a relatively low potential, near sodium plating. Hard carbon, however, has issues including volume expansion during the cycle and a comparatively low electrical conductivity [212]. This feature restricts the rate capability of HC because to safety concerns since overpotentials from electrode kinetics would cause the undesirable metallic plating (at high charging rates). The first cycle irreversible capacity loss in HC anodes, however, is the most pressing unresolved problem [213].

A large number of sodium ions are either consumed in the SEI production or trapped in the microstructure after the first cycle. Since the cathode material is the source of the cations and the anode's capacity must be greater than the cathode's in order to prevent metallic Na plating, this creates a dilemma for cell balance. This results in a significant decrease in the energy density of HC-based SIBs since the anodic capacity loss cannot be made up for by permitting an excess capacity in the cathode. Amorphous carbon exhibits strong cycling performance, high reversible specific capacity, good salt storage capacity, and a clear commercialization trend. Simultaneously, both hard and soft carbon can enhance electrochemical performance through atom doping to create nanostructured carbon materials, decrease the sodium ion diffusion path, and increase the layer spacing of materials [214]. Hard carbon is currently being used as an anode material by more producers of sodium-ion batteries. Low sodium storage potential, moderate sodium storage capacity (greater than titanium materials, lower than alloy materials), minimal volume deformation following sodium embedding, strong cycle performance, and the greatest overall performance among numerous contenders are all features of amorphous carbon [215]. From the standpoint of raw materials, amorphous carbon precursors are readily prepared and originate from a variety of sources. Amorphous carbon is most likely to lead the way in the field of sodium-ion battery anode materials to attain industrialization when combined with the

forementioned benefits. Hard carbon materials are currently the most effective high specific energy battery anode materials for sodium-ion batteries, and their capacity and cycle performance are nearly in line with application requirements. However, the largest obstacle to the commercialization of these materials is how to lower production costs. Other carbon materials include carbon nanostructures like graphene. Carbon nanostructures can offer a lot of Na^+ adsorption sites due to their enormous surface area. In a same vein, they face challenges in addition to their distinct advantages [216]. Although graphene boasts a large surface area, exceptional strength and rigidity, excellent electrical and thermal conductivity, and ultra-thin and transparent properties, its manufacture is costly and its technology is still in its infancy [217]. Furthermore, theoretical studies show that monolayer graphene's Na^+ storage capacity is less than $20 \mu\text{A h cm}^{-2}$ and that the adsorption of ions, particularly Na^+ , is detrimental in terms of energy. It is difficult to achieve exact size control in green synthesis, but other carbon nano-materials can offer high surface areas and short diffusion distances to enable mass movement.

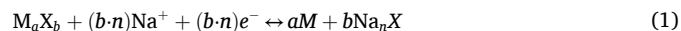
5.2. Alloy-Based anode materials

Alloying-type reaction mechanism materials, derived from groups IVA (e. g., Si, Ge, Sn, Pb) and VA (e. g., P, Sb, Bi) of the periodic table, are highly regarded as promising anode materials due to their high theoretical capacity, good conductivity, suitable working potential, and cost-effectiveness [218,219]. It has attracted a lot of interest from researchers and some progress has been made. However, these alloy anodes undergo a multi-electron transfer alloying reaction mechanism, leading to significant volume changes and voltage fluctuations during charge/discharge. This can cause voltage hysteresis, inactivation of active materials, instability of the solid electrolyte interface (SEI), and ultimately affect cycling performance [220]. However, the slow diffusion kinetics of Na^+/K^+ in alloy-type materials throughout the reaction process would restrict the electrode's rate performance. The low electrical conductivity of some particular elements, like P, is another important factor impeding their development. Researchers worked hard to solve these problems, including regulating the particle size, adding carbon materials, creating advantageous nanostructures, and preparing compatible electrolytes [221]. To address these challenges, nano-material design and hybridization with carbon or inert elements have been proposed to mitigate these issues, improving the discharge/charge process and minimizing volume effects. However, this approach may lead to a reduction in the overall energy density of the material, an

increase in SEI formation, and a decrease in initial Coulombic efficiency (ICE). In addition, the low tap density of nanostructures could impact the battery's bulk energy density. It is important to consider that well-designed nanomaterials necessitate intricate production processes, resulting in high manufacturing costs. In contrast, micro or bulk materials exhibit greater potential for practical applications across various aspects including volumetric energy density, Coulombic efficiency, manufacturing processes, and cost-effectiveness (Fig. 19) [222]. The advantages of micro or bulk alloy anodes include higher tap density, enabling increased mass loading and consequently enhancing volumetric capacity. Furthermore, the reduced specific surface area of these materials diminishes contact with the electrolyte, thereby improving the ICE in the first cycle [223]. The simplicity of manufacturing processes and the resulting cost advantages render micro or bulk materials appealing for practical applications.

5.3. Conversion anode material

Many metal oxides (MOx) and metal sulfides (MS_n), particularly layered metal disulfides (MS₂, M = Mo, Sn, W, and Ti), are anode materials based on conversion reactions. However, the conversion process is rarely exclusive during sodium insertion and is usually accompanied by alloying and/or intercalation activities [224]. The conversion reaction has greater theoretical potential than intercalation because of the multiple electron reaction that occurs during charge and discharge. During the uptake and extraction of Na⁺, conversion-type materials experience a phase transition that is accompanied by bond creation and breaking [225]. The conversion reactions involving Na⁺ can generally be expressed as follows:



Where X is the anionic species, n is the formal oxidation state of X, and M is a transition metal. Theoretical capacities for conversion reactions are two or three times greater than those of intercalation-type materials when the transition metal is completely reduced to a metallic state [226]. For instance, during the sodiation process, FeS₂, a common conversion-type anode, can be totally reduced by Na⁺ to create Na₂S and metallic Fe, exhibiting a high theoretical capacity of 894 mA h g⁻¹. Additionally, the electrochemical potential of the conversion-type anode can be modified by mixing different anionic species (X) and transition metal cations (M), thereby meeting the safety requirements of SIBs. Transition metal oxides, sulfides, selenides, and phosphides are examples of conversion-based anode materials that are being thoroughly studied for sodium storage [227]. The conversion reaction toward sodium's persistent issues with high voltage hysteresis, large volume change, slow reaction kinetics, and undesired electrolyte breakdown lead to low energy storage efficiency, poor cycling stability, poor rate

performance, and low initial Coulombic efficiency (ICE). However, transition metal oxides and phosphate materials, most of which are semiconductor materials, have a very limited capacity and still need to be modified through research to improve their reversible capacity. Compared with metal phosphide and metal oxide, the theoretical specific capacity of metal sulfide is lower, the voltage platform range is wider, and the volume expansion rate during the electrochemical cycle is much smaller than the latter two [228]. Therefore, the effect of volume expansion is still the main research direction of metal sulfides. Some researchers use a simple solvothermal synthesis of nitrogen-doped graphene sheet supported MoS₂ nanomaterial to enhance the conductivity of MoS₂ through loading. Reducing the size of MoS₂ nanomaterials shortens the Na⁺ transmission path and enlarges the layer spacing of MoS₂ to increase the Na⁺ transmission rate, which significantly improves the electrochemical performance of MoS₂. Compared with alloy anode material, the conversion anode material has a higher specific capacity, and the volume expansion rate in the electrocycling process is lower than that of alloy anode material. Conversion anode materials are often accompanied by intercalation reaction or alloying reaction during the conversion reaction, because the addition of metal increases the electrical conductivity [229]. According to the activity of the metal can be divided into active metal Sn and inactive metal gel two categories. The active metal will undergo a conversion reaction and alloying reaction with the alkali metal Na, and the resulting product has a high specific capacity, while reducing the volume expansion of the complex. In addition, in order to restrain the volume expansion of the material and cause the falling off the composite material, carbon-based materials are usually added for composite modification, and the excellent conductive network of carbon materials can also improve the transmission rate of Na⁺ [230].

5.4. Organic compounds

Organic chemicals are more attractive than inorganic electrodes because of their affordability, adaptability, and sustainability [231]. The majority of organic electrodes currently in use are carbonyl derivatives, usually Na₂C₈H₄O₄; biomolecular and Schiff-based compounds have also been discovered in recent years. Its unique embedding/ejection mechanism enables it to have a fast rate of sodium ion migration. However, due to the low intrinsic electron conductivity of most organic anode materials, many conductive additives need to be added in the pulping process, which reduces the bulk energy density of the battery system. At the same time, excessive conductive additives will have side reactions with the electrolyte at low potential, reducing the first coulomb efficiency [6]. Organic compounds are rich in chemical composition, low in cost, environmentally friendly, can achieve multi-electron reaction, electrochemical window can be adjusted, but there are defects such as poor electronic conductivity and easy to dissolve in

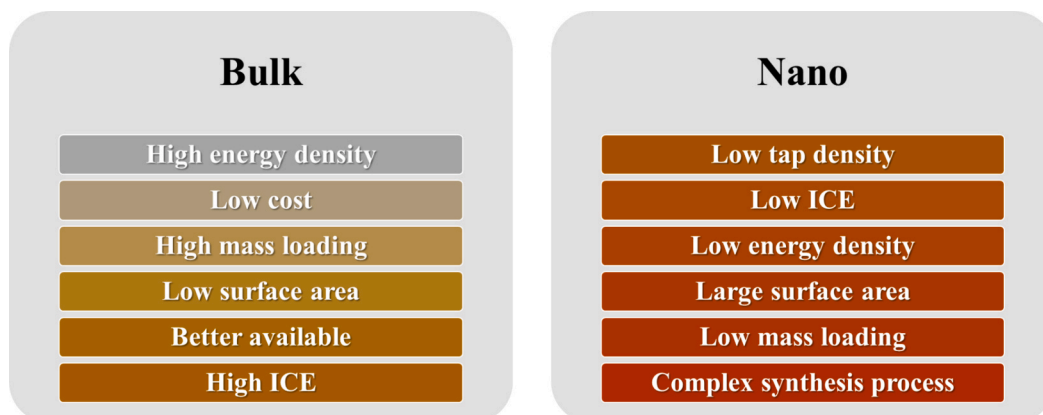


Fig. 19. Comparison of micron-sized and nanosized alloys.

electrolyte [232]. Organic compounds have been widely studied for their advantages of high theoretical capacity, low cost, resource sustainability, structural diversity, and safety [233]. Furthermore, because of their advantages biodegradability, abundance, light weight, and ease of processing natural-derived redox biomolecules are considered attractive options [234]. Wu S et al. introduced long-lasting organic electrodes with a stable α -C radical intermediate. The resonance effect and steric effects successfully suppress the unpaired electron's excessive reactivity, resulting in an electrode with stable cycling for over 2,000 cycles and 96.8% capacity retention. The α -radical also shows reversible transformation between C=C, α -C-radical, and α -C-anion. This change increases the electrodes' Na⁺ storage by more than 0.83 Na⁺ insertion per α -C radical. The purported process of TSAQ (tris N-salicylideneanthraquinoylamine) sodiation and desodiation presented in Fig. 20a. In contrast, a C[•] radical intermediate, DCCA-3Na, is also produced by the three-electron reduction of the dichloroisocyanuric acid (DCCA) electrode (Fig. 20b). However, due to a side reaction with the electrolyte or inside the radical molecule, DCCA-3Na instantly converts to electro-inactive DCCA-3Na-A. When designing organic electrodes with improved cycling life and energy storage capacity, the intermediate radical stabilization technique may be instructive.

5.5. MXenes materials

MXenes are a newly developed, large category of 2D transition metal carbides, carbonitrides, and nitrides that have garnered a lot of interest

lately [235]. The general composition of the MXenes family is $M_{n+1}X_nT_x$, where X is C and/or N, Tx stands for terminal surface groups –O, –F, and/or –OH, and M is a transition metal such as Ti, V, Mn, Mo, Nb, Cr, Sc, etc. This is usually prepared by selectively etching A layers, such as Al, Si, and Ga, from the corresponding $M_{n+1}AX_n$ phases.

MXenes were found to satisfy nearly every electrode/host material requirement [236]: The superior electrical conductivity enhances the kinetics of reactions by enabling quick ion and electron transit within structures; In certain batteries based on sodium chemistry, the metal-based species in the MXene phase chemically adsorb the hazardous intermediates, including polysulfides and polyselenides; MXene sheets' unique two-dimensional shape offers a perfect foundation for creating a range of composite structures with other materials, such as polymers, metal-based species, and carbonaceous materials; and Long-term stability in batteries based on sodium chemistry is made possible by the strong structural integrity [237].

The outstanding electrochemical results produced by batteries based on sodium chemistry are made possible by the synergistic effects of the aforementioned functions, opening the door for commercialization [238]. Because of their remarkable metallic conductivity, adjustable surface chemistry, and two-dimensional layered structure, MXenes have been considered as promising candidates for supercapacitors, rechargeable batteries, catalysts, oxygen reduction and evolution, water purification, electromagnetic interference shielding, pressure sensors, and field-effect transistors [239]. The use of MXene-based materials in sodium-ion storage has been extensively studied through theoretical

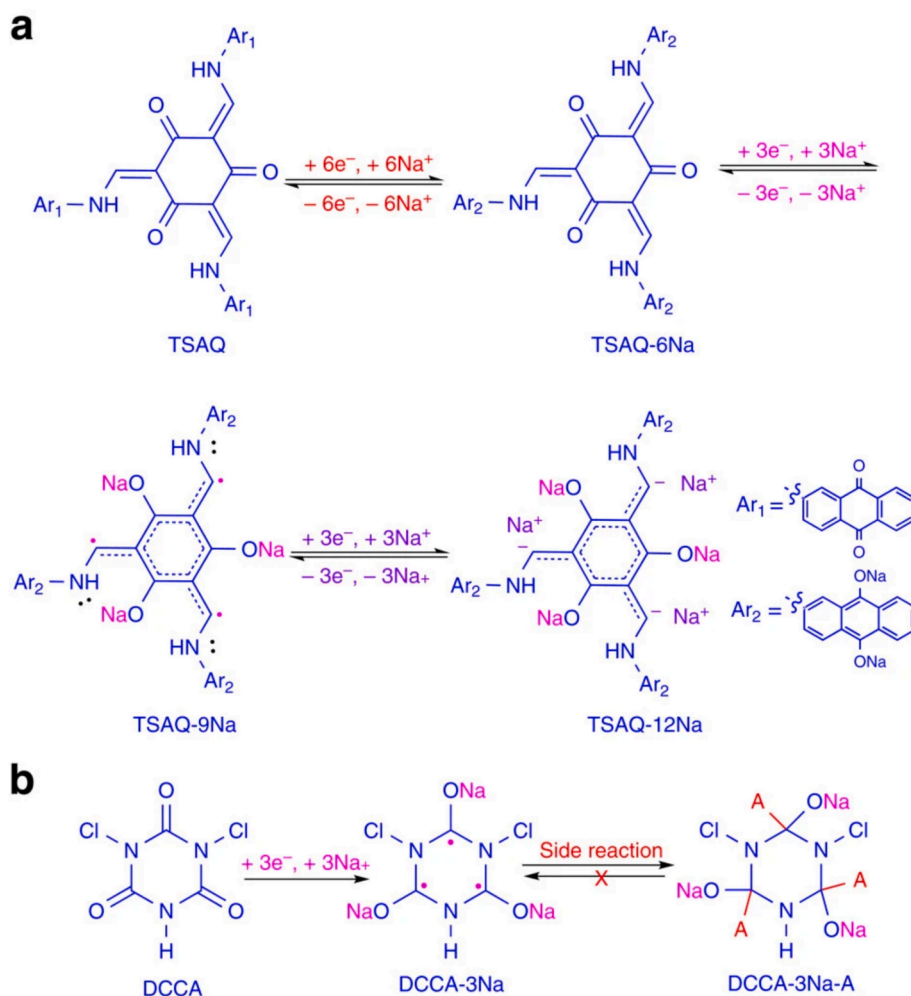


Fig. 20. (a) The TSAQ electrode's reversible electrochemical process and (b) the DCCA electrode's production of inactive material. Groups interacting with the radical intermediate are denoted by "A." Reproduced under a Creative Commons Attribution 4.0 International License [6]. Copyright 2016, Nature.

models and experimental observations. The findings reveal that materials based on MXene have a lot of promise as sodium-ion storage hosts. Additionally, they offer significant understanding of the sodium-ion storage mechanisms for materials based on MXene [240]. Since the structures of MXenes, such as surface groups and interlayer spacing, can govern their properties, many attempts have been made to optimize the electrochemical performance of MXene-based materials [241]. Ma P et al. used a one-pot self-assembly technique to create robust 2D VS₂/Ti₃C₂T_x MXene nanostructures with a strong Ti-S covalent bond [235]. With a remarkable sodium-ion storage capacity of 424.6 mAh g⁻¹ and an ultrahigh average initial coulombic efficiency of 95.08%, the SIBs based on the VS₂/MXene electrode demonstrate first-class electrochemical performance even at 10 A g⁻¹. It also exhibits a remarkable sodium-ion storage capacity of 514.2 mAh g⁻¹ at 1 A g⁻¹, with about 100% capacity retention after 500 high-rate cycles. Hence, SEM and TEM are used to measure the morphologies and microstructures of materials. The monodispersed and homogeneous VS₂ microflowers, as depicted in Fig. 21a, are made up of many nanosheets with a diameter of about 1.5 μm and a thickness of around 20–50 nm. Furthermore, a self-stacked, crinkled sheet morphology with an ultrathin thickness is clearly depicted in the MXene SEM image (Fig. 21b). As seen in Fig. 21c, MXene closely wraps the VS₂ nanosheets following the self-assembly process. In contrast to pure MXene (1.3 nm, Fig. 21e), the few-layer feature of MXene can be shown by TEM (Fig. 21d) with a greater layer spacing of 1.51 nm. Additionally, the VS₂/MXene HRTEM image (Fig. 21f) shows distinct lattice fringes with d-spacing of 2.51 and 2.30 Å, which correspond to the (011) plane of VS₂ (JCPDS No. 89–1640) and (104) plane of MXene, respectively. The presence of both nanosheets is further demonstrated by the SAED pattern (Fig. 21g), which shows the in-plane diffraction of planes for both VS₂ and MXene. Furthermore, the distribution of V, S, Ti, and C is seen in the STEM image (Fig. 21h) and matching EDS spectrum

(Fig. 21i), indicating the successful combination of VS₂ with MXene through the self-assembly process.

Generally, available and tested possible anode materials for SIB applications are summarized in Table 2.

6. Electrolytes

Electrolytes are another essential part of rechargeable batteries. Battery cycle life, Safety, rate capability, and are parameters that depend on the type of electrolyte. These factors also depend on the stability and compatibility of the solid-electrolyte interphase (SEI) that is generated. [262]. Liquid, solid, or semi-solid electrolyte phases are all possible. Electrochemical stability, chemical stability, ionic conductivity, thermal stability, electrical insulation and low toxicity are all requirements for electrolytes in SIBs that are essentially the same as those in LIBs. The use of volatile and extremely flammable organic electrolyte solvents is one of the primary sources of risk in SIBs; this problem becomes more severe in the case of the long-size stationary batteries required for renewable power sources. The limitations of these electrolytes are comparable to those discovered in LIBs earlier. Low vapour pressure organic solvents and both inorganic and organic solid electrolytes have been suggested as solutions to this issue [263]. Ionic liquids (ILs) are liquid, usually below 100 °C, and are made up of ions (both cations and anions), as their name indicates (Table 3). A high dielectric constant that promotes thermodynamic, non-flammability, and good electrochemical, ion mobility, chemical stabilities and non-volatility are all beneficial characteristics of ILs. Additionally, by selecting the right anionic and cationic components, these characteristics can be adjusted. In addition to these uses in practical electrochemistry, ionic liquids are becoming a potent type of organic solvent used in many different fields, such as phytochemical, analytical chemistry, drug transport and delivery, biomass treatment,

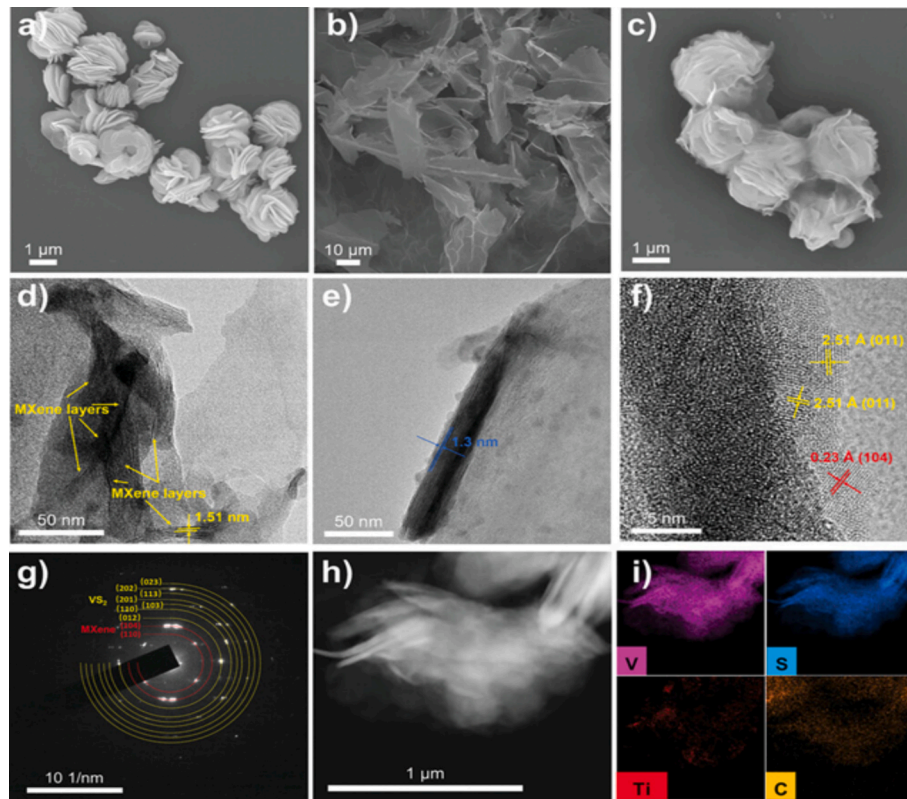


Fig. 21. A–c) sem images of VS₂, MXene, and VS₂/MXene. d,e) TEM images of VS₂/MXene and MXene. f,g) HRTEM image and the SAED pattern for VS₂/MXene hybrid. h,i) The matching EDS mapping of the distribution of V, S, Ti, and C elements and the STEM image of the VS₂/MXene hybrid. Reproduced under the terms of the Creative Commons CC BY license [235]. Copyright 2023, Wiley-VCH. Generally, available and tested possible anode materials for SIB applications are summarized in Table 2.

Table 2
Potential anode materials used for SIBs.

S. no.	Material	Electrochemical performance	Synthesis method	Morphology	Ref.
	Hierarchically porous carbon/graphene	After 1000 cycles, 250 mAh/g at 1000 mA/g	Ionothermal process	Sandwich-like	[242]
	N-doped carbon	After 200 cycles, 134 mAh/g at 200 mA/g	Pyrolyzation	Nanofibers	[243]
	Yolk-shell $\text{Sn}_4\text{P}_3\text{@C}$	After 400 cycles, 360 mAh/g at 1500 mA/g	Solution-based phosphorization treatment	Nanospheres	[244]
	TiO_2 /carbon	After 100 cycles, 140.4 mAh/g at 100 mA/g	Template method	Hollow spheres	[245]
	Porous nitrogen-doped carbon	After 600 cycles, 200 mAh/g at 200 mA/g	Template-assisted method	Sphere	[246]
	Hollow carbon	After 100 cycles, 200 mAh/g at 100 mA/g	Hydrothermal	Nanospheres	[247]
	Anatase TiO_2	After 1000 cycles, 100 mAh/g at 1680 A/g	Hydrolysis	Nanocubes	[248]
	$\text{Na}_2\text{Ti}_3\text{O}_7$	After 300 cycles, 110 mAh/g at 5C	Exfoliation process	Nanoplatelets and nanosheets	[249]
	Fe_2O_3	After 200 cycles, 1200 mAh/g at 800 mA/g	Hydrothermal	Multi-shelled core-shell microspheres	[250]
	Sn/Sb@porous carbon	After 200 cycles, 400 mAh/g at 500 mA/g	Electrospinning	Nanofiber	[251]
	MoS_2 /electrospun carbon	After 500 cycles, 200 mAh/g at 1000 mA/g	Chemical vapor deposition	Nanofiber	[252]
	$\text{Na}_2\text{Ti}_3\text{O}_7$	After 10,000 cycles, 65 mAh/g at 35C	Hydrothermal method	Nanoarrays	[253]
	3D MoS_2 -graphene	After 600 cycles, 322 mAh/g at 1500 mA/g	Spray pyrolysis process	Microspheres	[254]
	$\text{Na}_2\text{Ti}_3\text{O}_7/\text{Na}_2\text{Ti}_6\text{O}_{13}$	After 100 cycles, ~85 mAh/g at 20 mA/g After 4000 cycles, 19.45 mAh/g at 2 A/g	Hydrothermal	Rod-like	[255]
	P@carbon	After 100 cycles, 1586.2 mAh/g at 520 mA/g	Ball-milling	Nanotube	[256]
	Yolk-shell $\text{SnO}_2\text{@C}$	After 100 cycles, 928.9 mAh/g at 100 mA/g	Stöber method	Nanospheres	[257]
	$\text{Na}_2\text{Ti}_6\text{O}_{13}$	After 80 cycles, 160.7 mAh/g at 20 mA/g	Solid-state method	Rod-like	[258]
	Sb-C	After 400 cycles, 450 mAh/g at 200 mA/g	Electrospinning	Nanofiber	[259]
	$\text{Na}_2\text{Ti}_3\text{O}_7/\text{Na}_2\text{Ti}_6\text{O}_{13}$	After 100 cycles, 92 mAh/g at 5C	Solid-state	Nanorods	[260]
	Hollow P	After 600 cycles, 1000 mAh/g at 2600 mA/g	Solvothermal	Nanospheres	[261]

Table 3
Summary of list of liquid electrolytes used in SIBs.

	Anode	Cathode	Na Salt	Solvent	Na Salt	Molarity	Ref.
Aqueous	AC	PBA	NaClO_4	H_2O	NaClO_4	1	[273]
	$\text{NaTi}_2(\text{PO}_4)_3$	$\text{Na}_3\text{V}_{1.3}\text{Fe}_{0.5}\text{W}_{0.2}(\text{PO}_4)_3$	Na_2SO_4	H_2O	Na_2SO_4	1	[273]
	$\text{NaTi}_2(\text{PO}_4)_3$	$\text{Na}_2\text{MnFe}(\text{CN})_6$	NaTFSI	H_2O	NaTFSI	30	[274]
Non-aqueous Electrolytes	CNF	$\text{Na}_3\text{V}_2(\text{PO}_4)_3$	NaPF_6	Diglyme	NaPF_6	1	[275]
	rGO	Na	NaOTf	Diglyme	NaOTf	1	[276]
	HC	$\text{Na}_3\text{V}_2(\text{PO}_4)_3$	NaPF_6	EC:DEC	NaPF_6	1	[277]
	TiO_2	Na	NaOTf	Tetraglyme	NaOTf	1	[278]
	TiO_2	Na	NaClO_4	PC	NaClO_4	1	[279]
	Na	$\text{Na}_3\text{V}_2(\text{PO}_4)_3$	NaBF_4	Diglyme	NaBF_4	0.5	[280]
	SiC-Sb-C	Na	NaClO_4	EC:DEC	NaClO_4	1	[281]
	CNT/Na-IL	$\text{Na}_{0.67}\text{Mn}_{0.75}\text{Li}_{0.25}\text{O}_2$	NaPF_6	DME	NaPF_6	1	[282]
	HC	$\text{NaNi}_{0.5}\text{Mn}_{0.5}\text{O}_2$	NaClO_4	PC	NaClO_4	1	[283]
	MoS_2/C	Na	NaClO_4	EC:DEC	NaClO_4	1	[284]
	HC	Na	NaFSI	$\text{C}_3\text{mpyrFSI}$	NaFSI	0.5	[285]
	Na	$\text{Na}_{0.67}\text{Mn}_{0.67}\text{Ni}_{0.23}\text{Mg}_{0.1}\text{O}_2$	NaFSI	$\text{PyT}_{1.4}\text{FSI}$	NaFSI	1	[286]
	Na	FeF_3/C NFs	NaFSI	$\text{PyT}_{1.3}\text{FSI}$	NaFSI	1	[263]

and organic synthesis, agents [264]. To improve electrolytes in SIBs, many combinations of electrolyte ingredients, including solvents, additives, sodium salts, and their corresponding ratios, have been investigated. SIBs are safer because sodium salts have a higher melting point than lithium salts, which makes drying them easier and increases their thermal stability [265]. The fire resistance and general safety performance of SIBs are significantly influenced by the thermal stability of sodium salt. NaBF_4 , NaTFSI, NaFSI, NaClO_4 , NaPF_6 , and NaTf, are common sodium salts used in SIB electrolytes [266]. The development of SIBs depends on the choice of electrolyte solvents. Organic solvents, solid-state electrolytes, and aqueous electrolytes are the three types of solvents utilized in SIBs. Stability, non-toxicity, cost, and other considerations are among the requirements for solvents in SIBs. However, because they are volatile and combustible, organic solvent electrolytes have drawbacks that compromise the safety of sodium-ion batteries.

Researchers have looked into structural changes to organic solvents to reduce their flammability. These changes include end-group substitution, fluorination, and lengthening the chain. To reduce the flammability of the electrolyte solution, some research has looked into replacing organic electrolytes with solid-state and aqueous electrolytes in addition to changing organic electrolytes. Numerous organic solvents, including carbonates, ethers, esters, and others, are taken from LIBs and used in SIBs [267]. Ether-based and ester-based electrolytes are two of the most promising electrolytes for SIBs. Although ester-based electrolytes are utilized commercially in LIBs, ether cannot be employed in marketed LIBs due to structural disruption caused by ether Li^+ co-embedding in graphite. Although hard carbon (HC) is regarded to be a promising anode for commercial use in SIBs, the low-temperature and quick charging requirements are not met by the ester-based electrolyte's poor kinetics. Notably, HC has open angular microcrystallites and graphite-

like microcrystallites that allow for the highly reversible co-embedding of ether and Na^+ [268]. An electrolyte system for commercial batteries is usually made up of conductive salts combined with organic carbonates, such as diethylene and ethylene carbonate. However, carbonates are usually volatile and unstable liquids at high temperatures, which can lead to thermal runaway-related explosions that endanger safety. The generation of hazardous gasses like hydrogen fluoride during operation and electrode corrosion that shortens battery life are further problems associated with organic carbonates. These are strong arguments for looking at substitute electrolyte materials that have better operational safety while maintaining some of the superior electrochemical characteristics (high ionic conductivity) of organic carbonates. Because of their mechanical and thermal durability, solid-state electrolytes (ceramics, polymers, and gel polymers) may be utilized as substitute materials [269]. Unfortunately, at the electrode–electrolyte interface, they usually exhibit increased resistance and less ionic conductivity. It is crucial to remember that thermal abuse, in which the separator is directly melted by the high temperature, is the immediate cause of thermal runaway. Thermal runaways can be indirectly caused by mechanical and electrical misuse. Internal electrical abuse leads to dendritic growth, which can puncture the separator, whereas mechanical abuse involves the physical penetration of cells that shred the separator. It is important to note that, as Fig. 22 illustrates, short-circuiting is a prevalent issue in all of these misuse situations. In order to stop the chemical cycle before thermal runaway, an ideal electrolyte would have great mechanical and thermal stability. Enhancing the electrolyte's mechanical stability is one recommendation should be improved. Although this would be advantageous, creating an electrolyte that is mechanically resistant to tearing and punctures would probably endanger some of its electrochemical qualities. Enhancing the casing material's mechanical qualities is an alternate recommendation. Propylene carbonate (PC), dimethyl carbonate (DMC), ethyl carbonate (DEC), ethyl carbonate (EC), and their mixes are examples of carbon-based liquid electrolyte solvents that exhibit low initial CE and poor rate performance. Sodium salt with a large radius anion, which can be categorized as either organic or inorganic, is the main solute. The former includes sodium hexafluorophosphate, sodium perchlorate, and so on; the later is primarily composed of sodium fluorosulfonic acid salts, sodium fluoride sulfonimide salts, and so on. Electrolytic solutions of salts dissolved in solvents make up the majority of electrolytes, which are liquids at room temperature. A solvent is referred to be “protic” if it includes labile H^+ (such

as ethanol or water); otherwise, it is called “aprotic.” The electrolyte is an essential component of any particular electrochemical device, despite the fact that it is typically thought of as a passive element. In addition to the electrolyte's bulk characteristics, the interfaces that form between the electrolyte and the two electrodes frequently determine the battery's overall performance [270]. As a result, comprehending the structure and characteristics of the electrolyte/electrode interphase as well as the chemical makeup of electrolytes is essential to the development of SIBs and, in fact, forms the foundation of contemporary electrochemistry. A generally suitable electrolyte composition for SIBs has not yet been determined, in contrast to the Lithium Hexafluorophosphate (LiPF_6) salt dissolved in organic carbonate solvents, which is currently the standard electrolyte used in LIBs [271]. However, because the electrolyte has a significant impact on the energy density, cycling ability, reversible storage capacity, safety, and rate capability of future SIB, this problem offers a fantastic research opportunity. Luckily, a number of DFT and experimental optimization techniques have been used to identify the proper aprotic electrolyte composition in SIBs, and they are now a key component of the state-of-the-art.

Solid polymer electrolytes, inorganic solid electrolytes and composite solid polymer electrolytes are some of the different varieties of Solid-state electrolytes (SSEs). Each of these varieties has distinctive qualities and is currently being developed. These materials currently have issues including high interface impedance and low room temperature conductivity. It will therefore take some time for them to industrialize. Non-flammable electrolytes with uses across a wide temperature range are the main focus of electrolyte research [272]. Finding electrolytes with qualities that would help lower extra expenses related to the heating, fire safety, etc. of the area where the battery is stored is necessary for stationary SIB applications. Furthermore, the cost of the battery cannot be much increased by such materials. Because SSEs are a non-flammable kind of electrolyte, they reduce the possibility of fire dangers and improve the battery system's overall safety. This is because sodium is highly reactive and could cause a serious reaction with organic liquid electrolytes. SSEs have remarkable mechanical qualities, broad electrochemical stability windows, and great thermal stability. As a result, they can sustain greater voltages without breaking down, increasing energy density without sacrificing security. SSEs can function as a more effective barrier against dendrites than porous separators in liquid electrolytes since they are solid and can be punctured, resulting in a short circuit in the cell. SSEs have a longer lifespan and

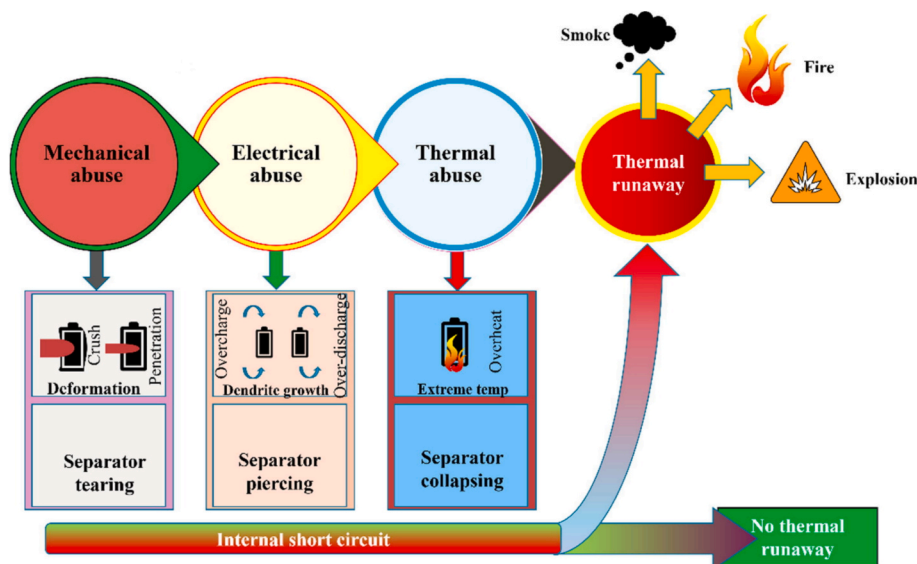


Fig. 22. Diagrammatic representation of the connection between abuse situations and internal short circuits. Reproduced under Creative Commons CC-BY license, with permission from [13]. Copyright © 2023, Elsevier.

require less maintenance than liquid electrolytes, but they are more expensive to manufacture and integrate. This and other features could be useful, particularly in fixed applications where battery weight and size are less important than in other application domains [262].

7. Separators

Separators can prevent short circuits, keep cathodes and anodes apart, and facilitate ion transport. The separator has a significant effect on the specific capacity and cycle stability of SIBs even if it is not the electrochemical active site that provides specific capacity [287]. To guarantee excellent performance, safety, and lifespan, a suitable separator material for SIBs must satisfy a number of crucial requirements. Good ionic conductivity to enable effective sodium ion movement, great mechanical stability to prevent short circuits, structural stability during battery operation, and excellent chemical resistance in reactive settings are all characteristics of ideal separator materials.

A wide operating temperature range, suitable thickness, superior thermostability at high temperature and electrochemical stability, excellent mechanical stability, porosity, the ability to avoid the formation and growth of dendrites, wettability and non-flammability are among the key qualities and properties of separators. Hence, separators affect the internal resistivity of the battery and affect its general performance [288]. Polymers, glass fiber, and cellulose-based materials are utilized in SIB separators. Although glass fiber materials are expensive and have little flexibility in battery applications, they offer good structural and thermal stability. Cellulose-based separators have the advantages of excellent wettability, environmental friendliness, high electrolyte uptake, degradability, and superior thermal stability, which make them promising material for SIB separators. Since cellulose is the most commonly used biopolymer (plant biomass contains 40–45 wt% cellulosic content), separators based on it are also available. With a wide range of uses, including energy storage systems, it is a particularly promising material in the field of renewable materials. Although cellulose-based separators are widely accessible and reasonably priced, their drawbacks include their flammability and moisture content (depending on the production process). Natural cellulose's abundant hydroxyl functional groups produce a powerful intermolecular interaction that decreases separator pore diameters and further affects Na^+ ion transport [289]. Polymeric materials are reasonably priced, have good cycle durability, and are structurally stable. However, because of their intricate chemical structures, they have low diffusion rates and ionic conductivity. Polyolefin-based materials are the most commonly utilized separator materials in SIB applications. Examples of materials in this group include glass fiber (GF), polypropylene (PP), and polyethylene (PE). These materials are typically utilized because of their low cost of large-scale manufacture and chemical stability. When utilized with carbon-based electrolytes, such as propylene carbonate (PC), ethylene carbonate (EC), and dimethyl carbonate (DMC), PP and PE have drawbacks such as poor wettability and inadequate electrolyte retention. The low temperature range of PP and PE, which have melting temperatures of 165 °C and 132 °C, respectively, is another disadvantage [290]. The material's expansion at higher temperatures poses the most risk to the battery because PP and PE tend to shrink, which can result in internal short circuits and other serious problems. The material's thickness and arrangement are not optimal for GF separators. Because of GF's high porosity, more electrolyte is required for filling, which increases the potential risk of dendritic development [291]. For its batteries, Aquion Energy, Inc., a business creating aqueous SIBs as a substitute for LIBs and other grid storage devices, uses a synthetic cotton separator and an electrolyte made of fire-safe seawater. However, because aqueous electrolyte cells have a narrow electrochemical stability window, each cell's output potential would be constrained. In order to adhere to recognized Li-ion cell production methods, Faradion Limited uses commercially widely utilized polyolefin separators and low-viscosity nonaqueous liquid electrolytes [292]. Regrettably, the

development of separators for SIBs has received far less attention than that of separators for LIBs. Because Na^+ and Li^+ have distinct kinetic properties, different materials are used even though LIBs and SIBs function on the same principle. Because LIB separators operate at similar temperatures and may employ electrolytes based on similar principles (e.g., salts in solvent LiPF_6 for Li-ion, NaPF_6 for Na-ion, etc.), it would be beneficial to investigate their appropriateness in SIBs in the future. Future advancements in separator material research may result from the combination of materials with complementary advantages. Therefore, combining the flexibility and stability of polymeric materials with the improved conductivity of nanomaterials could result in a separator material that is stable, flexible, and has good conductivity at a reasonable cost, supporting the development of commercially viable SIB separators [293].

8. Perspectives

SIBs are popular due to their inexpensive cost, large specific energy, long cyclic life, high safety, robust rate performance, and fast charging. The following research directions and optimization methodologies are essential to develop SIBs with high performance.

- **Structure-effect relationship:** It is essential to increase our understanding of the structure-effect relationship in electrode materials and how it affects electrochemical performance. Conventional modification techniques, such as the creation of carbon composites and heteroatom doping, are extensively studied. To obtain exceptional electrochemical performance at low-temperature (LT), however, novel approaches are desperately needed. We still don't fully comprehend the structure-effect relationship, despite this requirement.
- **Theoretical mechanisms development:** Although new energy technologies have advanced in recent decades, our scientific understanding has not advanced significantly. Electrochemistry itself has not witnessed much conceptual innovation, despite the advancement of energy storage materials and the emergence of new concepts. It is still difficult to comprehend the dynamic nature of batteries and the electrode–electrolyte interaction. There is currently no cohesive system-level theory for electrochemistry. In the future, incorporating computer technology could aid in the development of more creative theoretical frameworks.
- **Comprehensive knowledge of the Na storage mechanism:** Research on the electrodes' in situ structure and component evolution as well as the formation of a cathode electrolyte interface (CEI) film during cycling is crucial for illuminating the Na storage mechanism and directing the development of high-performance cathode materials. Thus, there should be a greater use of sophisticated characterisation methods including in situ electrochemical monitoring, in situ X-ray absorption spectroscopy (XAS), and in situ neutron diffraction.
- **Developing materials with high specific capacity, high redox potential, good structural stability, and reasonable electron and ionic conductivity is essential for cathode materials.** Environmentally friendly cathode materials and abundant, affordable resources are also essential. By doping and surface modification, the specific energy with stable structure can be effectively increased. Additionally, the development of cobalt-free cathodes is becoming popular as a way to cut expenses and emissions.
- **Large voltage windows and large temperature ranges must be developed for the electrolytes.** To build high-voltage SIBs, the electrochemical window and temperature ranges of the electrolyte can be expanded by developing additional functional molecules and adjusting electrolyte components.
- **Developing novel materials for electrodes and electrolytes:** Battery technology, particularly SIBs, can be transformed by new materials. Material research is a major factor in battery growth. For example, fast reactions, great capacity, and long-lasting performance can be

guaranteed by materials with many active sites and good mass transfer. Ion and electron mobility in LT circumstances may be improved by precisely arranging materials at a microscopic or even mesoscale utilizing “precision chemistry.” Ideally, materials with low energy barriers for electron and ion transfer might be used everywhere in these kinds of settings. High-entropy compounds have recently emerged as a prominent topic for research. Certain characteristics are displayed by the high-entropy compounds. Electronic conductivity, heat/air stability, and structural stability are all improved by the higher entropy. Furthermore, inorganic–organic hybrid materials show promise as SIB electrode materials. Ionic liquid electrolytes and solid polymer electrolytes have drawn a lot of interest. The next generation of sodium ionic electrolytes is thought to be ionic liquid electrolytes.

- An in-depth examination of reaction pathways in conjunction with sophisticated in situ characterisation techniques and theoretical computation. A well-defined atomic or molecular reaction mechanism greatly enhances the overall performance of SIBs. Studying the charge and discharge mechanism using sophisticated in situ characterization techniques has become popular with the advancement of modern in situ characterization techniques. Furthermore, theoretical calculations are an effective way to analyze reaction mechanisms. Theoretical calculations can be used to determine binding energy, adsorption energy, migration energy barrier, and other factors that are crucial for understanding the reaction mechanism.
- Developing large-scale synthesis techniques: Due to their straightforward compositions and available raw materials, anode materials like silicon and hard carbon can be produced in large quantities with high consistency. In contrast, the synthesis of cathodic materials, which contain complex components and costly raw materials (such as compositions based on Cu, Co and V), typically requires laborious synthesis steps, making the mass production of cathode materials extremely difficult. The coprecipitation method and solid-state ball milling are examples of synthesis procedures that can be expanded to large-scale manufacturing with enhanced batch stability and product yields.
- Optimizing the construction of SIBs: The absence of an alloying reaction between Na and Al is the basis of the bipolar electrode design, which can help accomplish effective recycling of electrode materials by using cheap aluminum as a shared current collector. Higher specific (volumetric) energy density, superior high-rate performance, and decreased cell resistance are just a few benefits of building SIBs with a bipolar electrode. However, electrolyte leakage and subsequent intermixing could lead to the batteries' exhaustive failure. There is still more research to be done on the validity and dependability of various battery designs.

9. Conclusion

In conclusion, the abundance and uniform geographic distribution of sodium resources make SIBs a technically viable and increasingly attractive to substitute LIBs in large-scale, cost-sensitive energy storage applications. Hence, SIBs are particularly attractive for grid-scale storage and application in areas with restricted access to supply chains for lithium.

There are still significant technological limitations in spite of these benefits. Because sodium has a larger ionic radius and a higher atomic mass, SIBs have an intrinsically lower energy density, both volumetric and gravimetric. In comparison to lithium, these characteristics lower specific capacity and packing efficiency. However, innovation in materials is making a big difference. This review summarizes the research advances of key components for SIBs, with a primary focus on recent developments in electrode materials. Hence, cathode materials used for SIBs such as transition metal oxides, polyanion compounds, and Prussian blue (analogues) are critically reviewed. Because of their delicate crystal structures, which are often resolved by lattice regulation, the

transition metal oxides are easily synthesized and exhibit high specific capacity and good rate capability. However, they are prone to collapse after repetitive Na insertion/extraction. Because of their strong X–O polar bonds and stable crystal structure, polyanionic materials offer great thermal/cyclic stability and high working voltages, but they have worse electronic conductivity. Although Prussian blue and its analogues are inexpensive, have excellent rate performance, and have changeable operating voltages, their chemical and structural instability is caused by the obstinate crystal water. Additionally, recent developments in anode materials for SIBs including alloy-based, intercalation-based, conversion-type, organic, and MXene materials are discussed. Carbon-based, alloy, and organic anode materials are used in sodium-ion batteries to provide superior cycle stability and storage capacity. However, these materials have issues that need to be addressed, like large volume expansion and poor electrical conductivity. An attractive group of materials are conversion-type anode materials, or the electrode materials utilized for conversion processes in LIBs or SIBs. Electrolytes used in SIBs can be liquid, solid, or semi-solid, and they should have low toxicity, strong ionic conductivity, and good mechanical and thermal stability. Battery safety can be improved by choosing the right electrolyte. Nevertheless, existing electrolytes are vulnerable to oxygen-releasing oxidation processes, which could lead to battery explosions. Because it stops short circuits between the cathode and anode, separator selection is also crucial for enhancing battery safety. Strong chemical resistance and superior heat stability are essential for the perfect separation. However, low ionic conductivity, poor ion diffusion rate, high prices, and restricted flexibility are common problems with current separators.

LIBs systems can be replaced by electrochemical systems based on Na ions. To make SIBs feasible in large-scale applications, research into electrode materials, electrolyte, current collectors, binders, carbon additives, and solvents is crucial. We think that increasing knowledge and discoveries about materials might accelerate the development of SIBs that are compatible with LIBs at the commercial level.

CRedit authorship contribution statement

Ababay Ketema Worku: Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Delele Worku Ayele:** . **Shiming Zhang:** . **Nithyadharseni Palaniyandy:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Bhekia B. Mamba:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Abolanle S. Adekunle:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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